

A risk-averse distributionally robust optimisation approach for drone-supported relief facility location problem[☆]

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ARTICLE INFO

Keywords:

Drone-supported last-mile humanitarian logistics
Drone-supported relief facility location problem (DSRFLP)
Distributionally robust optimisation (DRO)
Ambiguity set

ABSTRACT

Drone-supported last-mile humanitarian logistics applications play a crucial role in the rapid and efficient delivery of essential relief items, such as medicine, blood, and vaccines, during disaster and emergency situations. This paper explores a novel drone-supported relief facility location problem (DSRFLP) aimed at establishing an effective drone-supported last-mile humanitarian logistics system. The problem involves making joint decisions for both pre-disaster and post-disaster phases while considering the characteristics of drone-based delivery operations and uncertain demands. In the pre-disaster phase, we make the decisions regarding the locations of drone-supported relief facilities, inventory prepositioning of relief items, assignment of drones to the opened facilities, and allocation of drones to disaster demand sites. In the post-disaster phase, we make decisions related to delivery quantities. To tackle the challenge of incomplete demand distribution information in chaotic disaster environments, we establish a distributionally robust optimisation (DRO) model to handle the uncertainty of demands. This model adopts worst-case mean Conditional Value-at-Risk as the risk measurement, reflecting the risk-averse attitude of humanitarian organisers. In this paper, three ambiguity sets (box, ellipsoidal, and polyhedral) are considered to describe the ambiguity distributions of demands. To overcome the computational challenge, we reformulate the DRO model under three ambiguity sets into two mixed-integer linear programming models and one second-order cone programming model, which can be efficiently solved by off-the-shelf solvers. Furthermore, we validate our proposed DRO model through a small-scale example and a large-scale case study based on the Lushan earthquake in China. The computational outcomes underscore the superior performance of the proposed DRO model to mitigate the impact arising from incomplete probability distributions. We propose managerial implications and insights to support the decision-making of humanitarian organisations based on the experimental results. Finally, we propose two extended models to incorporate multiple relief items and equity constraints in priority settings and conduct numerical experiments to adapt to various real-world disaster scenarios.

[☆] The work described in this paper was supported by the National Natural Science Foundation of China (Grant number: 72301229) and Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hong Kong SAR (RJJ9, RJ85).

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<https://doi.org/10.1016/j.tre.2024.103538>

Received 18 December 2023; Received in revised form 5 March 2024; Accepted 11 April 2024

Available online 8 May 2024

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1. Introduction

The loss of countless lives and widespread devastation in disaster-prone areas due to the occurrence of natural and human-caused disasters have been common across the globe in the past few years. Based on statistics from reputable authorities, there were 387 instances of natural disasters around the world in 2022, resulting in economic losses of USD 223.8 billion and claiming 30,704 lives, with 185 million individuals affected by disasters in terms of economic loss, livelihood, physical injury, danger of death and home damage (CRED, 2023). The significant quantity of casualties can be attributed to the lack of prompt rescue and timely provision of emergency aid resources (Zhang et al., 2021). Therefore, it is crucial to establish a well-organised humanitarian logistics system to enhance efficiency in terms of preparations for disaster relief and response phases in disaster management.

The last-mile distribution phase plays a crucial role in the overall humanitarian logistics system, serving as a pivotal stage in facilitating disaster relief efforts. However, orchestrating the logistics arrangement for this stage consistently presents a formidable challenge (Noyan et al., 2016). For instance, in geophysical and hydrological disasters, such as earthquakes, the initial impact and subsequent aftershocks can lead to crush injuries and severe bleeding. In biological disasters, the spread of airborne and vector-borne diseases becomes a concern if sanitation conditions are extremely poor or there is a shortage of first-aid resources (Kouadio et al., 2012). In these disaster scenarios, characterised by scarce resources and challenging environmental conditions, a diverse range of essential relief items is imperative. Responding effectively to these multifaceted challenges requires a well-coordinated and rapid deployment of relief items to disaster sites. Failure in the immediate distribution of emergency relief items may exacerbate the victims' health conditions and result in a higher fatality rate. Hence, the establishment of a high-efficiency last-mile humanitarian logistics system becomes paramount for ensuring the timely distribution of these crucial relief items after disaster strikes.

The last-mile humanitarian logistics planning faces various challenges (Kovács and Spens, 2009). One of the major obstacles is the vulnerability of ground transportation networks in disaster areas to disruptions (Akbari et al., 2021). The extensive damage to the physical transportation infrastructure is unforeseeable in terms of its nature, scale, and extent, which might impede the last-mile humanitarian logistics for critical relief items. For example, a magnitude-9.0 earthquake that happened on March 11, 2011, in Japan generated a giant tsunami, and the natural disaster destroyed massive coastal infrastructures, which seriously affected the effectiveness and accessibility of transportation and rescue work. It was estimated that approximately 29 railway stations, 78 bridges, and 4000 road segments were damaged in this incident (Zhang et al., 2021). Due to the lack of essential transportation infrastructure, many disaster areas are inaccessible by ground-based transportation. The process of clearing routes to these areas often entails substantial time and effort, resulting in a missed opportunity for optimal rescue operations. Consequently, there is a pressing need to explore alternative transportation means to boost the efficiency of last-mile humanitarian logistics.

The rapid progress in unmanned aerial systems (UAS) technology, as well as guidance, positioning, and navigation systems, has led to the emergence of unmanned aerial vehicles (UAVs), commonly known as drones. Drones are capable of fulfilling multiple roles, such as urban traffic monitoring, package delivery, surveillance, and remote sensing (Barmounakis and Geroliminis, 2020; Gu et al., 2023; He et al., 2022; Kim et al., 2023; McCarthy et al., 2021). Furthermore, drones offer promising applications in disaster relief management. Firstly, drones deliver emergency supplies to disaster areas by air, which can reach locations that are inaccessible to traditional ground vehicles. Secondly, drones have faster mobility compared to manned ground vehicles. It can be deployed in a quick manner during the relief process. In 2023, after the occurrence of an earthquake of magnitude 7.8 that devastated some regions of Turkey and Syria, several drones were deployed to help with the transportation of emergency medicines, supplies, and food for victims supported by the National Disaster Response Force (Kandavel, 2023).

However, there are also many challenges when using drones to serve last-mile humanitarian logistics. For drone-based delivery operations, several characteristics set this operational setting apart from traditional truck-based delivery operations. Notably, drone deliveries might transport only a limited number of items per trip over short distances, constrained by payload weight limitations and battery capacity requirements. This differs significantly from conventional truck-based delivery operations, which typically involve longer distances and larger quantities of items (Chen et al., 2021; Zhu et al., 2022; Bruni et al., 2023). To address the challenge of limited flying distance and payload weight, a valid measure to deal with this problem is to set up drone-supported relief facilities, where drones and relief items can be assigned before a disaster, and delivered by drones to various disaster sites after the strike of a disaster (Zhu et al., 2022). Hence, this study proposes a drone-supported relief facility location problem (DSRFLP) given payload weight limitations and battery capacity requirements, which can address the challenge and promote the efficient operations of drone-supported last-mile humanitarian logistics.

Generally, uncertainty is an important characteristic that cannot be ignored, with demand uncertainty standing out as a critical factor in humanitarian logistics. The importance of managing demand uncertainty is underscored by the fact that a severe supply shortage could lead to ineffective rescue efforts and an increase in casualties (Bozorgi-Amiri et al., 2012; Zhong et al., 2020). The demand information from the disaster demand sites significantly influences decisions regarding the locations of drone-supported relief facilities. However, acquiring complete demand information is challenging due to the complexities of precisely evaluating disaster situations in affected regions. To address this challenge, a distributionally robust optimisation (DRO) approach is proposed for the DSRFLP to capture the ambiguous demand information of the disaster demand sites. Furthermore, in chaotic disaster relief systems, decision makers are often risk-averse rather than adopting a risk-neutral stance (Wang et al., 2023a). Conventional risk-neutral models may produce suboptimal results when applied to specific instances of random data, potentially leading to undesirable outcomes (Elçi and Noyan, 2018). Therefore, it becomes imperative to account for the risk-averse attitudes of decision makers and employ an appropriate risk measurement to mitigate risks in drone-supported last-mile humanitarian logistics. Conditional Value-at-Risk (CVaR) stands out as a widely utilised risk measurement. However, it primarily concentrates on the mean loss above a specified confidence level, neglecting the portion below the confidence level (Guan et al., 2024). To address this limitation, scholars

propose to introduce a weight parameter to integrate the expectation with CVaR, named mean-CVaR (MCVaR) (Wang et al., 2021; Guan et al., 2022). Moreover, traditional CVaR is commonly applied based on the assumption of a known probability distribution for uncertain parameters, but it often lacks robustness (Wang et al., 2023a). To cope with this challenge, the worst-case CVaR (WCVaR) has gained increasing attention (Zhu and Fukushima, 2009). Hence, we establish a risk-averse DRO model incorporating the worst-case MCVaR (WMCVaR) as the risk measurement to solve the DSRFLP in this paper. Currently, there is some literature that integrates the DRO approach and risk preference measures (Chang et al., 2017; Wang et al., 2021). In essence, this combination enables the DRO model to tackle challenges associated with incomplete probability distributions of random parameters and hedge against uncertainty to avoid high-risk solutions (Feng et al., 2023).

More specifically, we address the DSRFLP under the uncertainty of disaster site demand within a two-stage decision framework. In the first stage, we determine the pre-disaster decisions, including the locations of drone-supported relief facilities, inventory prepositioning of relief items, drone assignment to the opened facilities, and drone allocation to disaster demand sites. This aligns with real-world practices, where certain higher-level decisions, such as those related to location planning and inventory, must be pre-determined and kept fixed to prevent significant disruptions in the pre-disaster phase. Moreover, in disaster-stricken environments, it is crucial to determine the drone visit schedule in the pre-disaster phase because humanitarian organisers need to communicate in advance to the managers of both disaster sites and low-altitude air traffic to ensure adequate preparation of the necessary workforce and equipment. Moreover, determining the drone visit schedule in advance is crucial for the planning of drone aerial paths, especially considering the complexities of geological and meteorological environments. This proactive approach ensures rapid and effective relief support in the post-disaster phase. Therefore, in this paper, the decision regarding the drone allocations to disaster demand sites is considered in the first stage, involving the coordination of transportation tasks for each drone to access its designated disaster demand sites. Importantly, this decision can be conducted independently of the demand realisations in the pre-disaster phase. Existing literature has addressed the decisions associated with vehicle visits in the first stage in the field of transportation and logistics planning (Adulyasak et al., 2015; Tricoire et al., 2012). Particularly in disaster relief scenarios, the advance determinations regarding drone visits assume more paramount significance due to the imperative of ensuring a prompt and efficient response. After the demand information becomes known, subsequent decisions regarding the delivery quantity can be made in the post-disaster phase.

Additionally, we propose two extended models to adapt to different real-world disaster scenarios, building upon the mathematical model of DSRFLP. One model aims to consider multiple relief items, encompassing diverse emergency needs such as medical kits, blood, and vaccines. This approach ensures that a diverse range of urgent needs can be met promptly and efficiently. The other model extension incorporates an equity constraint in priority setting. We prioritise relief resource allocation according to the urgent degree and incorporate equity constraints to ensure a fair distribution of resources among affected populations. This approach promotes the efficiency and efficacy of drone-supported last-mile humanitarian logistics but also reflects principles of social justice and enhances social welfare.

To sum up, the main contributions of this research are summarised as follows:

(1) We concentrate on a practical DSRFLP that incorporates joint decisions during pre-disaster and post-disaster phases, including drone-supported relief facility location, inventory prepositioning of relief items, drone assignment to the opened facilities, drone allocation to disaster demand sites, and delivery quantity decisions. To encompass the specific features of drone-based delivery operations, we incorporate considerations of battery capacity requirements and payload weight limitations into this problem. Additionally, we consider the uncertainty of the demands of disaster demand sites. This study is the first attempt to integrate these decisions within the field of drone-supported last-mile humanitarian logistics management.

(2) Considering the difficulty of acquiring the complete demand information in disaster scenarios, we propose a two-stage DRO model for DSRFLP to capture the ambiguous distribution information of the uncertain demand. Additionally, WMCVaR risk measurement is adopted in this DRO model to reflect the decision maker's risk-averse attitude. This is the first study to adopt the risk-averse DRO approach to make hybrid decisions under partial probability distribution regarding uncertain demands in the field of drone-supported last-mile humanitarian logistics.

(3) Three ambiguity sets (box, ellipsoidal, and polyhedral) are designed to depict the partial probability distributions of the demand. Furthermore, we reformulate the DRO model under three ambiguity sets into the corresponding equivalent computationally tractable formulations. Additionally, we conduct a series of experiments using a small-scale example and a large-scale case study based on the Lushan Earthquake to verify the effectiveness of the DRO model under three ambiguity sets. The results of the numerical experiments demonstrate that the DRO model exhibits better performance than the stochastic programming model, which presents a compelling motivation for relief organisers to embrace the DRO approach. Finally, we provide several valuable managerial implications and insights, serving as a valuable reference for the decision-making processes of humanitarian organisers.

(4) We propose two extended problems of DSRFLP to consider multiple relief items and equity constraint in priority setting. For these problems, we first present the two-stage stochastic programming models and then establish the corresponding DRO models under box, ellipsoidal, and polyhedral ambiguity sets to address the demand uncertainty. Moreover, we perform numerical experiments to validate the performance of the proposed extended model, thereby providing valuable insights for optimising disaster relief efforts and adapting to various real-world disaster scenarios.

The remainder of this paper is organised as follows. Section 2 conducts a comprehensive review of prior work in the related fields, identifying key research gaps. In Section 3, we investigate a risk-averse two-stage stochastic programming and introduce a DRO model within three ambiguity sets (box, ellipsoidal, and polyhedral). Moreover, we transform the DRO model, considering different ambiguity sets, into an equivalent computationally tractable formulation suitable for resolution with a commercial solver. Section 4 details various numerical experiments conducted to validate the efficacy of the DRO model and presents several managerial insights.

In Section 5, we introduce two extended models aimed at incorporating multiple relief items and equity constraint in priority setting, establishing upon the foundation of the DSRFLP. Additionally, we conduct numerical experiments to evaluate the performance of the proposed extended models. Finally, Section 6 provides conclusions and outlines potential avenues for future research.

2. Literature review

2.1. Humanitarian logistics planning under uncertainty

In humanitarian logistics, it is crucial to deliver relief items to disaster sites as quickly and securely as possible. However, there are lots of inherent uncertain factors that will have a significant influence on the delivery process. As stated by Tofighi et al. (2016), the inherent uncertainties cannot be neglected because they affect the efficiency of the humanitarian logistics system. Therefore, many scholars focus on humanitarian logistics planning considering various uncertain factors aiming at designing an efficient and rapid-response system to save lives and support people in need.

Scholars have widely adopted stochastic programming as a methodology to address humanitarian logistics planning problems, given the inherent significance of uncertainty. In the stochastic programming model, the precise distributions of the uncertain parameters are known perfectly. For instance, Shu et al. (2023) established a chance-constrained stochastic programming model to obtain the facility location, inventory pre-positioning, and the assignment of relief facility to disaster areas considering the uncertainty of demand. Particularly, the joint chance constraint is employed to assess the responsiveness of the humanitarian relief network. Rath et al. (2016) introduced a stochastic programming model to obtain the optimal decisions of the intermediate depot locations and vehicle acquisition considering the uncertainty of accessibility of the road network in disaster relief operations. Hu et al. (2017) investigated a stochastic model to optimise the pre-disaster inventory quantity and post-disaster procurement planning considering demand uncertainty. Manopiniwes and Irohara (2017) established a multi-objective model to optimise the relief operations during the preparedness and response phases based on the stochastic programming framework. Hu et al. (2019) considered heterogeneous vehicles and incorporated the uncertainty of road network state into a two-stage stochastic programming framework to solve a relief distribution problem. The uncertain road state was represented by dynamic road capacity. Wang and Nie (2022) integrated pre-disaster decisions regarding emergency supply and transportation network mitigation, along with post-disaster decisions involving dynamical transportation planning considering traffic congestion effects based on the stochastic programming approach.

The aforementioned literature has assumed the distributions of uncertain parameters are known perfectly. However, this may not be possible in most real-world scenarios. Aiming at this problem, scholars proposed a robust optimisation method to avoid the negative impacts on the decision-making process due to a lack of accurate information on uncertain parameter distribution (Bertsimas and Sim, 2004; Gorissen et al., 2015). Hence, there is much literature that focuses on obtaining optimal decisions in humanitarian logistics based on the robust optimisation method. For instance, Ni et al. (2018) proposed a min-max robust model to determine the facility location, inventory pre-positioning, and delivery operations considering the uncertainties of the proportions of the useable inventories, demand, and road capacities. Yahyaei and Bozorgi-Amiri (2019) designed a network for relief logistics considering the facility disruption risks in order to distribute resources timely and hedge against uncertain environments based on the robust optimisation approach. Avishan et al. (2023) presented an adaptable robust optimisation model to address a humanitarian logistics problem considering uncertain travel time. This model made joint decisions regarding routing and the service times of the disaster sites for relief logistics teams. Furthermore, Sun et al. (2021) introduced a robust optimisation model to minimise total cost while also reducing injury severity scores.

The empirical distribution estimated from the historical data is always partial and inaccurate in the stochastic programming, while the decisions obtained from the robust optimisation are always too conservative due to a lack of distribution information of the uncertain parameter. To address the limitations of stochastic programming and robust optimisation, DRO has emerged as a promising methodology, which offers a solution to handle the uncertainty without the need for complete distribution information of the uncertain parameter (Delage and Ye, 2010). Additionally, this methodology can mitigate the overly conservative nature of the solutions obtained from the robust optimisation method (Goh and Sim, 2010). In the field of humanitarian logistics planning, this methodology has been gradually applied to make some crucial decisions. Zhang et al. (2022) investigated a DRO model to minimise total cost and time while simultaneously maximising the equitable distribution of relief supplies. Additionally, a DRO model was established by Wang et al. (2022) to address a relief logistics problem considering lateral transshipment strategy. Subsequently, Wang et al. (2023a) applied a DRO approach to decide the relief facility location, inventory quantity, and delivery planning. Yang et al. (2023) addressed a location-allocation problem in humanitarian logistics considering the dynamic relief facility location and relief distribution based on the DRO approach. Additionally, Shehadeh and Tucker (2022) proposed a DRO model to obtain optimal locations and inventory prepositioning of opened warehouses and joint decisions on the procurement and distribution decisions of relief items.

2.2. Drone-supported relief operation

Drones are one of the most powerful and promising novel technologies to boost the efficiency of disaster response and relief operation management (Rejeb et al., 2021). Recently, more and more scholars have focused on the optimisation of drone-supported relief operations. For instance, Zhang et al. (2021) investigated an optimisation model to make routing decisions in a collaborated truck-and-drone system to provide precise delivery of humanitarian operations. Rabta et al. (2018) investigated an optimisation

model for drone-based last-mile relief distribution with the installation of charging stations. Wang et al. (2023c) developed a multi-objective drone-based queuing-location model considering uncertain demand and congestion. Additionally, Kim et al. (2019) investigated an optimisation model to decide facility location and drone delivery assignments across multiple time periods to effectively cope with disasters. Additionally, this study accounted for the uncertain flight distance of drones through a chance constraint. Ghelichi et al. (2022) introduced a chance-constrained programming model to obtain optimal drone launching platform locations under the demand uncertainty of disaster-affected areas.

Noted that recent research was conducted by Zhu et al. (2022) that focused on the robust facility location problem with drones in a relief scenario. The paper aimed at locating relief facilities and deploying drones to deliver to customers considering the uncertainty of demands based on a robust optimisation framework. We conclude three major differences from this research as follows. First, we consider the hybrid decisions of facility location, inventory of the relief items, drone assignment, drone visit schedule, and delivery quantities. The decisions are more comprehensive, considering the pre-disaster and post-disaster phases simultaneously. Second, this paper applies the DRO approach to solve the problem, which is distinct from the conservative robust optimisation model proposed by this literature. Third, we adopt the WMCVaR into our DRO model to reflect the risk-averse attitude of the relief organisers.

To the best of our knowledge, there is currently a lack of existing literature that pertains to the DSRFLP based on the DRO approach. Therefore, this study contributes to the research agenda by addressing the following specific gaps. First, no previous research has made integrated decisions regarding drone-supported relief facility location, the inventory of the relief items, drone assignment to the relief facility and allocation to the disaster demand sites, and delivery quantities given battery and payload weight requirements in the field of drone-supported last-mile humanitarian logistics planning. Second, the application of the DRO approach in this field is limited. Current research in this field is unavailable to address the conservative nature of robust optimisation and the influence of uncertain distribution information in stochastic programming. Third, within the context of drone-supported last-mile humanitarian logistics planning, there is a lack of literature that considers the risk-averse attitudes of decision makers to hedge the potential risk of the system. To fill these gaps, we introduce a novel risk-averse DRO model adopting the WMCVaR risk criterion under three ambiguity sets, namely box, ellipsoidal, and polyhedral, to address the DSRFLP to obtain the joint decisions.

3. Problem description and mathematical formulations

3.1. Problem description

In contemporary disaster relief scenarios, there has been a notable surge in the utilisation of drones. This escalating trend underscores the growing prominence and adoption of drone technology in the humanitarian sector. The diverse capabilities of drones, encompassing efficient delivery of essential supplies (medicine, blood, vaccines, etc.) to remote or inaccessible areas, significantly contribute to enhancing the overall effectiveness and responsiveness of disaster relief efforts. In drone-supported last-mile humanitarian logistics, the drone-supported relief facilities play a pivotal role in furnishing essential logistical support and services. However, the number of open drone-supported relief facilities might be limited because of geographical and practical factors. Therefore, in this study, we focus on a practical DSRFLP aimed at optimising the planning of the drone-supported last-mile humanitarian logistics system.

We present a comprehensive overview of the DSRFLP and subsequently propose a two-stage stochastic programming model. For this problem, our modelling framework incorporates decision-making in both the pre-disaster and post-disaster phases. First, we outline the sets, decision variables, and parameters in the first stage (pre-disaster). During the pre-disaster phase, our focus spans the determination of optimal locations for drone-supported relief facilities, inventory prepositioning of relief items, drone assignment to the opened facilities, and drone allocation to disaster demand sites. In this problem, the drone-supported relief facility functions both as a warehouse for storing relief items and as a distribution hub where a limited number of drones can pick up items destined for the disaster demand sites and change the battery. We let K and L denote the set of candidate drone-supported relief facilities and the set of capacity levels of the open drone-supported relief facilities, respectively. For each $k \in K$ and $l \in L$, we introduce the binary variable x_{kl} that equals 1 if a drone-supported relief facility is open at location k with a designated capacity level l , and is 0 otherwise. The unit opening cost for a drone-supported relief facility with the capacity level l is represented as a non-negative parameter c_l^{OPEN} . Let D denote the set of available drones, and each available drone $d \in D$ can be assigned to an opened drone-supported relief facility to transport the relief items to the disaster demand sites. For each $d \in D$ and $k \in K$, we define binary variable y_{dk} that equals 1 if the drone d is assigned to the open drone-supported relief facility k , and is 0 otherwise. As mentioned earlier, our approach involves determining the allocation of drones to disaster demand sites during the pre-disaster phase, ensuring a swift response in the post-disaster phase. Here, we let I represent the set of disaster demand sites. Hence, for each $i \in I$, $d \in D$, and $k \in K$, we define z_{idk} that equals 1 if disaster demand site i is visited by drone d assigned to the drone-supported relief facility k , and is 0 otherwise. In this study, each assigned drone launches from the drone-supported relief facility while carrying a parcel travels to the disaster demand site, delivers the relief items, and returns to the drone-supported relief facility. This entire transportation process is termed a *drone trip*. For drone operation research, battery capacity is a key element that needs to be considered compared to the traditional truck-based delivery operations (Kim et al., 2019; Hong et al., 2018; Chen et al., 2021). Therefore, we introduce the maximum flying range B^{MAX} resulting from the limited battery capacity to meet the requirements of battery capacity of drone-based delivery operations. Drones are permitted to make multiple trips if their battery capacity allows, as in prior studies (Zhu et al., 2022; Chauhan et al., 2019). Fig. 1 shows an illustrative example of a drone-supported last-mile humanitarian logistics system. In this system, there are two open drone-supported relief facilities F1–F2 and 12 disaster demand sites S1–S12. Six drones are assigned to provide relief delivery services for the disaster sites. We note that drones can make several one-to-one drone trips (open drone-supported relief

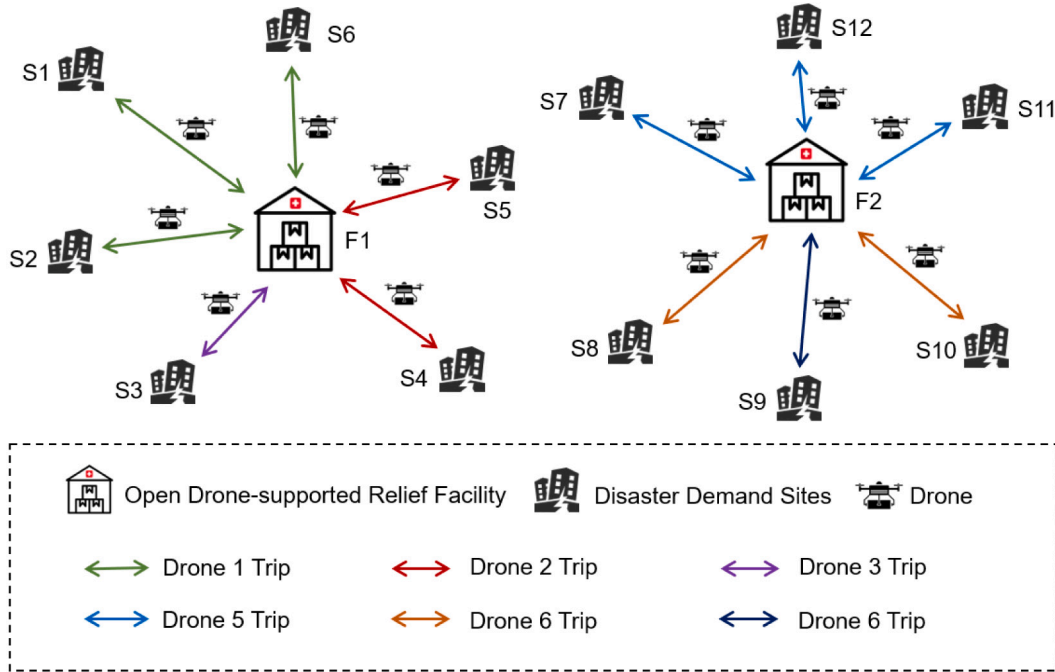


Fig. 1. An illustrative example of a drone-supported last-mile humanitarian logistics system.

facility to disaster demand site and back) until the battery is depleted. For instance, drone 1 completes three drone trips to serve disaster demand sites S1, S2, and S6, while drone 2 makes two drone trips to serve disaster demand sites S4 and S5. Once the drone visit schedules are established, the subsequent step involves planning and managing the aerial paths in the pre-disaster phase. This process could incur drone aerial path management costs, which might be associated with path planning and the deployment of manpower or equipment to manage low-altitude airspace. Therefore, we define a non-negative parameter c_{ki}^{MANA} as drone aerial path management cost from the drone-supported relief facility k to the disaster demand site i . Additionally, in our consideration, we focus on a singular type of relief supply, where critical items (such as blood, medical kits, and pharmaceuticals) can be provided in the bundled form (Elçi and Noyan, 2018). For each $k \in K$, we define the non-negative continuous decision variable v_k as the inventory prepositioning quantity of relief items in the drone-supported relief facility k . For the open drone-supported relief facility with capacity level l , it has limited inventory capacity of relief items represented by W_l and drone capacity denoted by Q_l . The unit inventory prepositioning cost for relief items is represented as a non-negative parameter c^{INV} .

Furthermore, we illustrate the sets, decision variables, and parameters in the second stage (post-disaster). Transitioning to the post-disaster phase, our model encompasses the critical decision-making process related to the delivery quantity of relief items to the disaster demand sites. We let Ξ represent the set of scenarios and $\xi \in \Xi$. Regarding the uncertain parameter, we let D_i^ξ indicate the demand of relief item for the disaster demand site i under scenario ξ . For each $i \in I$, $k \in K$, and $\xi \in \Xi$, we introduce the non-negative continuous decision variable w_{ik}^ξ , representing the number of relief items delivered to disaster demand site i from drone-supported relief facility k under scenario ξ . We also consider the determination of the delivered quantity of individual drones in this paper. For each $i \in I$, $d \in D$, $k \in K$, and $\xi \in \Xi$, we let the non-negative continuous decision variable q_{idk}^ξ represent the number of relief items delivered to disaster demand site i by the drone d that is assigned to drone-supported relief facility k under scenario ξ . Note that another special characteristic of drone delivery operations that needs to be considered is that drones may only deliver a small number of items per trip due to payload weight limitations, in contrast to traditional truck-based delivery operations (Chen et al., 2021; Golabi and Nejad, 2022). Therefore, we denote the maximum payload weight limit of drones as L^{MAX} to constrain the quantity of relief items transported by drones per drone trip. The delivery cost for unit demand from the drone-supported relief facility k to the disaster demand site i is represented by the non-negative parameter c_{ki}^{DELI} . Finally, for each $i \in I$ and $\xi \in \Xi$, the non-negative continuous decision variable u_i^ξ is introduced, which means the quantity of unsatisfied demand of disaster demand site i for the relief items under scenario ξ . And for each $k \in K$ and $\xi \in \Xi$, we let the non-negative continuous decision variable r_k^ξ express the quantity of unused inventory of the relief items at the drone-supported relief facility k under scenario ξ . The penalty cost for per unit unsatisfied demand of relief items and unused relief items are represented by the non-negative parameters c^{UNS} and c^{UNU} , respectively. We summarise the notations in Table 1.

Table 1

Notations.

Notation	Definition
Sets	
I	The set of disaster demand sites, $i \in I$.
K	The set of candidate drone-supported relief facilities, $k \in K$.
L	The set of capacity levels of drone-supported relief facilities, $l \in L$.
D	The set of available drones, $d \in D$.
Ξ	The set of scenarios, $\xi \in \Xi$.
Parameters	
c_l^{OPEN}	The unit opening cost for a drone-supported relief facility with capacity level l .
c_{kl}^{MANA}	The drone aerial path management cost from the drone-supported relief facility k to the disaster demand site i .
c^{INV}	The unit inventory prepositioning cost for relief items.
c_{ki}^{DELI}	The delivery cost for unit demand from the drone-supported relief facility k to the disaster demand site i .
c^{UNS}	The penalty cost for per unit unsatisfied demand of relief items.
c^{UNU}	The penalty cost for per unit unused relief items.
Q_l	The capacity of drones for the drone-supported relief facility with capacity level l .
W_l	The inventory capacity of relief items for the drone-supported relief facility with capacity level l .
B^{MAX}	The maximum flying range of drones due to the limited battery capacity.
L^{MAX}	The maximum payload weight limit of drones.
D_i^ξ	The demand of relief item for the disaster demand site i under scenario ξ .
Variables	
First stage variables	
x_{kl}	Binary variable, if the candidate location k is chosen as drone-supported relief facility with capacity level l , it equals to 1; otherwise, it equals to 0.
y_{dk}	Binary variable, if the drone d is assigned to the drone-supported relief facility k , it equals to 1; otherwise, it equals to 0.
z_{idk}	Binary variable, if the disaster demand site i is visited by drone d assigned to the drone-supported relief facility k , it equals to 1; otherwise, it equals to 0.
v_k	Continuous variable, the inventory prepositioning quantity of relief items in the drone-supported relief facility k .
Second stage variables	
w_{ik}^ξ	Continuous variable, the number of relief items delivered to disaster demand site i from drone-supported relief facility k under scenario ξ .
q_{idk}^ξ	Continuous variable, the number of relief items delivered to disaster demand site i by the drone d that is assigned to drone-supported relief facility k under scenario ξ .
u_i^ξ	Continuous variable, the quantity of unsatisfied demand of disaster demand site i for the relief items under scenario ξ .
r_k^ξ	Continuous variable, the quantity of unused inventory of the relief items at the drone-supported relief facility k under scenario ξ .

3.2. Risk-averse two-stage stochastic programming

In this subsection, the risk-averse two-stage stochastic programming model for the DSRFLP is presented, which relies on the perfect distribution information of the random demand.

We first introduce the risk-neutral two-stage stochastic programming model. In the first stage, we focus on the pre-disaster decisions regarding the drone-supported relief facility locations, inventory prepositioning of relief items, drone assignment to the opened facilities, and drone allocation to disaster demand sites. Our primary objective is to minimise the total cost, which encompasses the opening cost of drone-supported relief facilities, the inventory prepositioning cost of relief items, and the drone aerial path management cost. The objective function can be formulated as follows:

$$\min \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} \quad (1)$$

The constraints in the first stage are expressed as follows:

$$\sum_{l \in L} x_{kl} \leq 1, \quad \forall k \in K, \quad (2)$$

$$y_{dk} \leq \sum_{l \in L} x_{kl}, \quad \forall d \in D, \forall k \in K, \quad (3)$$

$$\sum_{k \in K} y_{dk} \leq 1, \quad \forall d \in D, \quad (4)$$

$$\sum_{d \in D} y_{dk} \leq \sum_{l \in L} Q_l x_{kl}, \quad \forall k \in K, \quad (5)$$

$$v_k \leq \sum_{l \in L} W_l x_{kl}, \quad \forall k \in K, \quad (6)$$

$$z_{idk} \leq y_{dk}, \quad \forall k \in K, \forall i \in I, \forall d \in D, \quad (7)$$

$$\sum_{k \in K} \sum_{d \in D} z_{idk} \geq 1, \quad \forall i \in I, \quad (8)$$

$$\sum_{i \in I} d_{ki} z_{idk} \leq B^{\text{MAX}} y_{dk}, \quad \forall k \in K, \forall d \in D, \quad (9)$$

$$x_{kl} \in \{0, 1\}, \quad \forall k \in K, \forall l \in L, \quad (10)$$

$$y_{dk} \in \{0, 1\}, \quad \forall d \in D, \forall k \in K, \quad (11)$$

$$z_{idk} \in \{0, 1\}, \quad \forall k \in K, \forall i \in I, \forall d \in D, \quad (12)$$

$$v_k \geq 0, \quad \forall k \in K. \quad (13)$$

Constraints (2) ensure that at most one capacity level can be chosen for each candidate drone-supported relief facility. Constraints (3) forbid drones to be assigned to a candidate drone-supported relief facility that is not open. Constraints (4) force each drone can only be assigned to an open drone-supported relief facility once at most. Constraints (5) guarantee the number of assigned drones cannot exceed the capacity of drones for each open drone-supported relief facility. Constraints (6) impose an inventory capacity limit on the quantity of inventory prepositioning for relief items at each open drone-supported relief facility. Constraints (7) state that no relief delivery service can be provided if a drone is not assigned to one certain drone-supported relief facility. Constraints (8) require each disaster demand site to be visited by drones at least once to prevent a severe shortage of relief items. Constraints (9) enforce battery range constraints on each drone d assigned to drone-supported relief facility k . Finally, constraints (10)–(13) define the domains of first stage variables.

In the second stage, we consider the post-disaster delivery quantity decisions in response to different disaster scenarios, which are made after the demand information becomes known. The objective of the second stage is to minimise the delivery cost as well as the penalty cost of unsatisfied demand and unused relief items. Given a set of the first stage decisions $\{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}\}$ and the random input data vector $\boldsymbol{\theta}(\xi)$, the recourse problem, defined as $Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))$ can be expressed as follows:

$$Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) = \min \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{DELI}} q_{idk}^{\xi} + \sum_{i \in I} c^{\text{UNS}} u_i^{\xi} + \sum_{k \in K} c^{\text{UNU}} r_k^{\xi} \quad (14)$$

$$\text{s.t.} \quad \sum_{k \in K} w_{ik}^{\xi} + u_i^{\xi} = D_i^{\xi}, \quad \forall i \in I, \quad (15)$$

$$\sum_{i \in I} w_{ik}^{\xi} + r_k^{\xi} = v_k, \quad \forall k \in K, \quad (16)$$

$$w_{ik}^{\xi} - \sum_{d \in D} q_{idk}^{\xi} = 0, \quad \forall k \in K, \forall i \in I, \quad (17)$$

$$q_{idk}^{\xi} \leq L^{\text{MAX}} z_{idk}, \quad \forall k \in K, \forall i \in I, \forall d \in D, \quad (18)$$

$$w_{ik}^{\xi} \geq 0, \quad \forall k \in K, \forall i \in I, \quad (19)$$

$$q_{idk}^{\xi} \geq 0, \quad \forall k \in K, \forall i \in I, \forall d \in D, \quad (20)$$

$$u_i^{\xi} \geq 0, \quad \forall i \in I, \quad (21)$$

$$r_k^{\xi} \geq 0, \quad \forall k \in K. \quad (22)$$

Formula (14) is the objective function of the second stage under scenario ξ associated with the transportation of the relief items to the disaster sites as well as the penalties of unsatisfied demand and unused inventory. Constraints (15) give a balance formulation, that is, the number of delivered relief items plus the quantity of unsatisfied demand is equivalent to the corresponding demand for each disaster demand site. Constraints (16) mean the number of relief items delivered from the relief facility plus unused inventory equals the corresponding inventory prepositioning quantity. Constraints (17) indicate the relationship between the two variables. The number of delivered relief items for individual drones can be calculated based on this equation. Constraints (18) require the number of relief items delivered by the drone cannot exceed the maximum payload weight limit. Ultimately, constraints (19)–(22) specify the domains of second stage variables.

Hence, the risk-neutral two-stage stochastic programming model for the DSRFLP is given as follows:

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}} \quad & \left\{ \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \mathbb{E}[Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))] \right\} \\ \text{s.t.} \quad & \text{Constraints (2)–(13), Constraints (15)–(22),} \end{aligned} \quad (23)$$

where $\boldsymbol{\Psi}$ denotes the vector composed by the variables of the second stage, i.e., $\boldsymbol{\Psi} = \{\mathbf{w}, \mathbf{q}, \mathbf{u}, \mathbf{r}\}$.

Due to the infrequency of catastrophic events, it is necessary to incorporate the risk consideration in the humanitarian logistics (Wang et al., 2023b). In the two-stage stochastic programming model, we adopt the MCVaR criterion to capture the risk preferences of decision makers. This is because integrating the expected cost and CVaR (i.e., MCVaR) helps mitigate the inefficiencies of both the risk-neutral approach and uncertainty risk simultaneously (Noyan, 2012). Subsequently, we elaborate on the risk-averse two-stage stochastic programming model for the DSRFLP. Some definitions are introduced before the establishment of the model.

Definition 1. The Conditional Value-at-Risk (CVaR) for the random variable Z at a confidence level $\alpha \in [0, 1]$ is defined (Rockafellar and Uryasev, 2002) as

$$\text{CVaR}_\alpha(Z) := \inf_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1-\alpha} \mathbb{E}[(Z - \eta)_+] \right\}, \quad (24)$$

where $(a)_+ := \max\{a, 0\}$ for $a \in \mathbb{R}$, and $\mathbb{E}$ is the expected value operator. η represents the value of the maximum loss suffered by the function $Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))$ when the confidence level is α .

Definition 2. The mean-CVaR (MCVaR) for the random variable Z at a confidence level $\alpha \in [0, 1]$ and a weight parameter $\lambda \in [0, 1]$ is defined (Luan et al., 2022) as

$$\text{MCVaR}(Z) := (1 - \lambda)\mathbb{E}(Z) + \lambda\text{CVaR}_\alpha(Z). \quad (25)$$

Remark 1. Note that the relative significance of the risk term will increase when weight parameter λ increases and so would lead to more risk-averse decisions.

Remark 2. Suppose that Z is a discrete random variable with realisations Z^ξ and corresponding probabilities p^ξ for scenario $\xi \in \Xi$, the expression (25) can be represented as the following linear program for finite probability spaces by introducing the auxiliary variable s^ξ (Mahmutoğullari et al., 2018).

$$\text{MCVaR}(Z) = \min(1 - \lambda) \sum_{\xi \in \Xi} p^\xi Z^\xi + \lambda \left(\eta + \frac{1}{1 - \alpha} \sum_{\xi \in \Xi} p^\xi s^\xi \right) \quad (26)$$

$$\text{s.t. } s^\xi \geq Z^\xi - \eta, \quad \forall \xi \in \Xi, \quad (27)$$

$$s^\xi \geq 0, \quad \forall \xi \in \Xi. \quad (28)$$

We observe that the MCVaR risk measure incorporates information about the expected value and CVaR of a random variable. As the values of α or λ increase, decision makers tend to become more risk-averse.

Leveraging these definitions, we propose the risk-averse two-stage stochastic programming model [SP-DSRFLP] as follows:

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi, \eta} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} \\ & + (1 - \lambda) \sum_{\xi \in \Xi} p^\xi Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) + \lambda \left(\eta + \frac{1}{1 - \alpha} \sum_{\xi \in \Xi} p^\xi s^\xi \right) \end{aligned} \quad (29)$$

$$\text{s.t. } s^\xi \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \eta, \quad \forall \xi \in \Xi, \quad (30)$$

$$s^\xi \geq 0, \quad \forall \xi \in \Xi. \quad (31)$$

3.3. Distributionally robust optimisation model

In reality, it is challenging to obtain the actual probability distribution of the demand due to complex environmental factors in disaster areas. Hence, we assume the demand distribution information $\mathbf{p}$ is incomplete and belongs to the ambiguity set $\mathcal{P}$. Therefore, the DRO model [DRO-DSRFLP] based on WMCVaR can be formally established as follows:

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \text{WMCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))) \\ \text{s.t. } \quad & \text{Constraints (2)–(13), Constraints (15)–(22),} \end{aligned} \quad (32)$$

where $\text{WMCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))) = \sup_{\mathbf{p} \in \mathcal{P}} \text{MCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)))$ expresses worst-case MCVaR of the second stage cost under ambiguity set $\mathcal{P}$. In this formula, $\sup_{\mathbf{p} \in \mathcal{P}}(\cdot)$ denote the worst-case scenario, i.e., min-max robust criterion. The objective (32) minimises the total cost of the first stage cost, the expected cost of the second stage, and the MCVaR of the worst distribution from the ambiguity set $\mathcal{P}$. Based on the definition of the MCVaR (Definition 2), we have $\sup_{\mathbf{p} \in \mathcal{P}} \text{MCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))) = (1 - \lambda) \sup_{\mathbf{p} \in \mathcal{P}} \mathbb{E}(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))) + \lambda \text{WCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)))$. Note that $\text{WCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)))$ denote the worst-case CVaR. Therefore, the above model can be transformed into as follows:

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} \\ & + (1 - \lambda) \sup_{\mathbf{p} \in \mathcal{P}} \mathbb{E}(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))) + \lambda \text{WCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))) \\ \text{s.t. } \quad & \text{Constraints (2)–(13), Constraints (15)–(22).} \end{aligned} \quad (33)$$

Assuming the uncertain parameter follows a discrete distribution, we can transform the model (33) as follows:

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} \\ & + (1 - \lambda) \max_{\mathbf{p} \in \mathcal{P}} Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{p} + \lambda \text{WCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))) \end{aligned} \quad (34)$$

s.t. Constraints (2)–(13), Constraints (15)–(22),

where $\mathbf{p} \in \mathcal{P}$ has finite support set Ξ , and $\mathbf{p} = \{p^1, p^2, \dots, p^{|\Xi|}\}^T$, in which $p^\xi > 0$ is the probability of the scenario $\xi \in \Xi$ and $\sum_{\xi \in \Xi} p^\xi = 1$.

This DRO model is an intractable semi-infinite programming model and lacks the amenability to leverage established optimisation techniques and commercial solvers for solving (Ling et al., 2017). Consequently, the need arises to transform the model [DRO-DSRFLP] into a equivalent computationally tractable formulation.

3.3.1. Ambiguity sets

Generally speaking, there are two principles that can be followed when choosing the ambiguity set. Firstly, the ambiguity set needs to harness the existing information pertaining to the distribution of uncertain parameters, i.e., a collection of historical data, which can be deemed as data samples extracted from the true distribution. Secondly, the ambiguity set needs to facilitate computationally tractable formulations (Guan et al., 2024). Based on the above principles, in this study, we address the partial information on the demand probability distribution of the disaster demand sites by constructing three ambiguity sets: box, ellipsoidal, and polyhedral. In the context of drone-supported last-mile humanitarian logistics planning, it is reasonable to consider these three ambiguity sets. Relief organisers commonly estimate the nominal demand distribution according to historical demand data. Moreover, to address the inexact distribution resulting from incomplete data, refinement can be achieved by introducing a disturbance term to improve accuracy. These three ambiguity sets have found applications in engineering and management fields, particularly, and have also been employed in addressing specific relief planning problems (Zhu and Fukushima, 2009; Guan et al., 2024; Qiu et al., 2014; Wang et al., 2023a, 2021). We define the box, ellipsoidal, and polyhedral ambiguity sets as the following definitions.

Definition 3. A box ambiguity set $\mathcal{P}^{\text{Box}}$, to which the distribution probability $\mathbf{p}$, can be defined as

$$\mathcal{P}^{\text{Box}} = \{\mathbf{p} \mid \mathbf{p} = \mathbf{p}^0 + \epsilon, \mathbf{e}^T \epsilon = 0, \|\epsilon\|_\infty \leq \omega\}, \quad (35)$$

where $\mathbf{e}$ is the unit column vector; $\mathbf{p}^0$ represents the nominal probability distribution vector of the uncertain demand, which means the most likely probability distribution; ϵ is the perturbation vector, $\epsilon = \{\epsilon_1, \epsilon_2, \dots, \epsilon_{|\Xi|}\}^T$; ω expresses the allowable bias limit of the perturbation vector ϵ , which can be adjusted and $\omega \in [0, 1]$. Note that $\mathbf{e}^T \epsilon = 0$ is necessary to ensure that the vector $\mathbf{p}$ can satisfy the basic requirements of the probability distribution.

Definition 4. An ellipsoidal ambiguity set $\mathcal{P}^{\text{Ellipsoidal}}$, to which the distribution probability $\mathbf{p}$, can be defined as

$$\mathcal{P}^{\text{Ellipsoidal}} = \{\mathbf{p} \mid \mathbf{p} = \mathbf{p}^0 + \mathbf{A}_1 \epsilon, \mathbf{e}^T \mathbf{A}_1 \epsilon = 0, \mathbf{p}^0 + \mathbf{A}_1 \epsilon \geq 0, \|\epsilon\|_2 \leq 1\}, \quad (36)$$

where $\mathbf{A}_1$ is the known scaling matrix of the ellipsoidal. ϵ expresses the uncertain parameter vector. The conditions $\mathbf{e}^T \mathbf{A}_1 \epsilon = 0$ and $\mathbf{p}^0 + \mathbf{A}_1 \epsilon \geq 0$ can ensure the satisfaction of the probability distribution requirements.

Definition 5. A polyhedral ambiguity set $\mathcal{P}^{\text{Polyhedral}}$, to which the distribution probability $\mathbf{p}$, can be defined as

$$\mathcal{P}^{\text{Polyhedral}} = \{\mathbf{p} \mid \mathbf{p} = \mathbf{p}^0 + \mathbf{A}_2 \epsilon, \mathbf{e}^T \mathbf{A}_2 \epsilon = 0, \mathbf{p}^0 + \mathbf{A}_2 \epsilon \geq 0, \|\epsilon\|_1 \leq 1\}, \quad (37)$$

where $\mathbf{A}_2$ is the known scaling matrix of the polyhedral. ϵ expresses the uncertain parameter vector. The conditions $\mathbf{e}^T \mathbf{A}_2 \epsilon = 0$ and $\mathbf{p}^0 + \mathbf{A}_2 \epsilon \geq 0$ can ensure the satisfaction of the probability distribution requirements.

3.3.2. Model equivalent reformulations

The DRO models mentioned above are computationally intractable. Therefore, we reformulate the proposed DRO model [DRO-DSRFLP] under three ambiguity sets into the equivalent reformulations that can be solved computationally.

Theorem 1. If the probability distribution vector $\mathbf{p}$ belongs to the box ambiguity set $\mathcal{P}^{\text{Box}}$, the model [DRO-DSRFLP] can be equivalently reformulated into the following mixed-integer linear programming model.

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi, \tau, \varsigma, \varrho, \hat{\tau}, \hat{\varsigma}, \hat{\varrho}, \mathbf{s}, \eta} & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\ & + (1 - \lambda) [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{p}^0 + \boldsymbol{\Omega}^T \varsigma + \boldsymbol{\Omega}^T \varrho] + \frac{\lambda}{1 - \alpha} [(\mathbf{p}^0)^T \mathbf{s} + \boldsymbol{\Omega}^T \hat{\varsigma} + \boldsymbol{\Omega}^T \hat{\varrho}] \\ \text{s.t.} & \mathbf{e} \tau + \varsigma - \varrho = Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)), \\ & \mathbf{e} \hat{\tau} + \hat{\varsigma} - \hat{\varrho} = \mathbf{s}, \\ & \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta \mathbf{e}, \\ & \varsigma \geq \mathbf{0}, \varrho \geq \mathbf{0}, \hat{\varsigma} \geq \mathbf{0}, \hat{\varrho} \geq \mathbf{0}, \mathbf{s} \geq \mathbf{0}, \\ & \text{Constraints (2)–(13), Constraints (15)–(22),} \end{aligned} \quad (38)$$

where $(\tau, \varsigma, \rho, \hat{\tau}, \hat{\varsigma}, \hat{\rho}) \in \mathbb{R} \times \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R} \times \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R}_+^{|\mathcal{I}|}$ are dual variables.

Proof of Theorem 1. The proof for Theorem 1 is given in Appendix A. Note that Proposition 1 is necessary during the proof process of Theorem 1, Proposition 1 and the corresponding proof can be found in Appendices B and C.

Theorem 2. If the probability distribution vector $\mathbf{p}$ belongs to the ellipsoidal ambiguity set $\mathcal{P}^{\text{Ellipsoidal}}$, the model [DRO-DSRFLP] can be equivalently reformulated into the following second-order cone programming model.

$$\begin{aligned}
 \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}, \boldsymbol{\theta}, \boldsymbol{\zeta}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\zeta}}, \boldsymbol{s}, \boldsymbol{\eta}} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\
 & + (1 - \lambda) [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))^T \mathbf{p}^0 + (\mathbf{p}^0)^T \boldsymbol{\theta} + \boldsymbol{\zeta}] + \frac{\lambda}{1 - \alpha} [(\mathbf{p}^0)^T \mathbf{s} + (\mathbf{p}^0)^T \hat{\boldsymbol{\theta}} + \hat{\boldsymbol{\zeta}}] \\
 \text{s.t.} \quad & \|\mathbf{A}_1^T \boldsymbol{\theta} + \mathbf{A}_1^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\theta} \mathbf{A}_1^T \mathbf{e}\|_2 \leq \boldsymbol{\zeta}, \\
 & \|\mathbf{A}_1^T \hat{\boldsymbol{\theta}} + \mathbf{A}_1^T \mathbf{s} - \hat{\boldsymbol{\theta}} \mathbf{A}_1^T \mathbf{e}\|_2 \leq \hat{\boldsymbol{\zeta}}, \\
 & \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\eta} \mathbf{e}, \\
 & \boldsymbol{\theta} \geq \mathbf{0}, \quad \boldsymbol{\zeta} \geq \mathbf{0}, \quad \hat{\boldsymbol{\theta}} \geq \mathbf{0}, \quad \hat{\boldsymbol{\zeta}} \geq \mathbf{0}, \quad \mathbf{s} \geq \mathbf{0}, \\
 & \text{Constraints (2)–(13), Constraints (15)–(22),}
 \end{aligned} \tag{39}$$

where $(\boldsymbol{\theta}, \boldsymbol{\zeta}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\zeta}}, \boldsymbol{s}) \in \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R} \times \mathbb{R}_+$ are Lagrange multipliers.

Proof of Theorem 2. The proof for Theorem 2 is given in Appendix D.

Theorem 3. If the probability distribution vector $\mathbf{p}$ belongs to the polyhedral ambiguity set $\mathcal{P}^{\text{Polyhedral}}$, the model [DRO-DSRFLP] can be equivalently reformulated into the following mixed-integer linear programming model.

$$\begin{aligned}
 \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}, \boldsymbol{\pi}, \boldsymbol{\kappa}, \boldsymbol{\varphi}, \hat{\boldsymbol{\pi}}, \hat{\boldsymbol{\kappa}}, \hat{\boldsymbol{\varphi}}, \boldsymbol{s}, \boldsymbol{\eta}} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\
 & + (1 - \lambda) [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))^T \mathbf{p}^0 + (\mathbf{p}^0)^T \boldsymbol{\pi} + \boldsymbol{\varphi}] + \frac{\lambda}{1 - \alpha} [(\mathbf{p}^0)^T \mathbf{s} + (\mathbf{p}^0)^T \hat{\boldsymbol{\pi}} + \hat{\boldsymbol{\varphi}}] \\
 \text{s.t.} \quad & \|\mathbf{A}_2^T \boldsymbol{\pi} + \mathbf{A}_2^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\kappa} \mathbf{A}_2^T \mathbf{e}\|_\infty \leq \boldsymbol{\varphi}, \\
 & \|\mathbf{A}_2^T \hat{\boldsymbol{\pi}} + \mathbf{A}_2^T \mathbf{s} - \hat{\boldsymbol{\kappa}} \mathbf{A}_2^T \mathbf{e}\|_\infty \leq \hat{\boldsymbol{\varphi}}, \\
 & \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\eta} \mathbf{e}, \\
 & \boldsymbol{\pi} \geq \mathbf{0}, \quad \boldsymbol{\varphi} \geq \mathbf{0}, \quad \hat{\boldsymbol{\pi}} \geq \mathbf{0}, \quad \hat{\boldsymbol{\varphi}} \geq \mathbf{0}, \quad \mathbf{s} \geq \mathbf{0}, \\
 & \text{Constraints (2)–(13), Constraints (15)–(22),}
 \end{aligned} \tag{40}$$

where $(\boldsymbol{\pi}, \boldsymbol{\kappa}, \boldsymbol{\varphi}, \hat{\boldsymbol{\pi}}, \hat{\boldsymbol{\kappa}}, \hat{\boldsymbol{\varphi}}) \in \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R} \times \mathbb{R}_+$ are Lagrange multipliers.

Proof of Theorem 3. The proof for Theorem 3 is given in Appendix E.

4. Numerical experiments

This section verifies the validity of the proposed model by conducting experiments on both a small-scale example and a large-scale case study. We utilise the Gurobi 10.0.0 commercial solver, executed on a personal computer equipped with an AMD Ryzen 9 7950X 16-Core Processor running at 4.50 GHz and 128.0 GB RAM, operating under Windows 10 (64-bit).

4.1. A small-scale example

4.1.1. Parameter settings

The proposed model specifically addresses DSRFLP to promote the relief operations of drone-supported last-mile humanitarian logistics, which generally have a much smaller service area compared to traditional trucks (Zhu et al., 2022). Therefore, publicly accessible datasets for the relief facility location problem are not appropriate for this specific research problem. To overcome this limitation, we randomly generate a small-case example specifically tailored to the characteristics of drone-supported relief operations to validate the performance of proposed models referring to the research conducted by Zhu et al. (2022).

This small-scale example involves 10 disaster demand sites S1–S10 and 5 candidate drone-supported relief facilities F1–F5. We consider three categories of drone-supported relief facilities with different capacity levels, i.e., small (S), medium (M), and large (L). Referring to the literature conducted by Zhu et al. (2022), the opening costs of small, medium, and large drone-supported relief facilities are set as 200, 300, and 500, respectively. The inventory capacities of relief items for the facilities are 50, 80, and 120, respectively. Furthermore, the capacities of drones for the facilities are 10, 15, and 25, respectively. In terms of the parameter settings related to relief items, the unit inventory prepositioning cost c^{INV} is assumed to be 140 (Shehadeh and Tucker, 2022). The unit delivery cost for unit demand from the drone-supported relief facility k to the disaster demand site i is proportional to the

Table 2

Euclidean distances between the disaster demand sites and the candidate drone-supported relief facilities.

	F1	F2	F3	F4	F5	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
F1	0	3.87	2.08	1.47	2.96	3.27	2.27	5.93	2.16	2.39	4.6	3.69	2.94	6.84	3
F2	3.87	0	5.9	4.53	5.78	6.41	5.23	7.38	5.3	5.76	7.58	7.42	1.1	3.43	4.38
F3	2.08	5.9	0	2.53	2.22	2	1.81	6.66	2.6	1.24	4.33	1.63	4.9	8.93	3.52
F4	1.47	4.53	2.53	0	4.16	4.28	3.5	4.51	0.81	3.38	3.25	4.08	3.85	6.95	4.47
F5	2.96	5.78	2.22	4.16	0	0.75	0.69	8.61	4.54	0.98	6.52	2.53	4.68	9.15	1.9
S1	3.27	6.41	2	4.28	0.75	0	1.18	8.62	4.53	0.9	6.3	1.88	5.31	9.74	2.64
S2	2.27	5.23	1.81	3.5	0.69	1.18	0	7.98	3.92	0.73	6.01	2.58	4.14	8.57	1.79
S3	5.93	7.38	6.66	4.51	8.61	8.62	7.98	0	4.09	7.75	3.15	7.83	7.3	7.92	8.92
S4	2.16	5.3	2.6	0.81	4.54	4.53	3.92	4.09	0	3.66	2.46	3.98	4.65	7.58	5.09
S5	2.39	5.76	1.24	3.38	0.98	0.9	0.73	7.75	3.66	0	5.56	1.86	4.68	9.02	2.52
S6	4.6	7.58	4.33	3.25	6.52	6.3	6.01	3.15	2.46	5.56	0	5.08	7.05	9.34	7.42
S7	3.69	7.42	1.63	4.08	2.53	1.88	2.58	7.83	3.98	1.86	5.08	0	6.38	10.53	4.34
S8	2.94	1.1	4.9	3.85	4.68	5.31	4.14	7.3	4.65	4.68	7.05	6.38	0	4.51	3.32
S9	6.84	3.43	8.93	6.95	9.15	9.74	8.57	7.92	7.58	9.02	9.34	10.53	4.51	0	7.81
S10	3	4.38	3.52	4.47	1.9	2.64	1.79	8.92	5.09	2.52	7.42	4.34	3.32	7.81	0

Table 3

Computational results of optimal objective values under the box ambiguity set.

	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$	$\lambda = 0.9$
$\alpha = 0.5$	54 478.56	54 943.49	55 395.16	55 823.35
$\alpha = 0.6$	54 735.38	55 314.23	55 885.25	56 456.27
$\alpha = 0.7$	54 986.24	55 802.68	56 609.78	57 416.87
$\alpha = 0.8$	55 422.82	56 361.12	57 183.15	57 987.26
$\alpha = 0.9$	55 490.69	56 396.71	57 288.49	58 180.26

Euclidean distance between two nodes, which can be expressed as $c_{ki}^{\text{DELI}} = 5 \times d_{ki}$, where d_{ki} represents the euclidean distance between the drone-supported relief facility k and the disaster demand site i (Zhu et al., 2022). The Euclidean distances between the disaster demand sites and drone-supported relief facilities are reported in Table 2. The flying speed of drones is 80 km/h. Additionally, we have a definition of shortage penalty coefficient ϵ to calculate the unit penalty cost per unit unsatisfied demand c^{UNS} . It equals the shortage penalty coefficient times the unit inventory prepositioning cost, which can be expressed as $c^{\text{UNS}} = \epsilon \times c^{\text{INV}}$ (Noyan, 2012). Regarding the penalty cost per unit unused relief items c^{UNS} , we supposed that it equals 1.2 times as expensive as the unit inventory prepositioning cost. As to the parameter settings related to drones, the drone aerial path management cost per unit flying time from the drone-supported relief facility to the disaster demand site is set as 2. According to the literature from Shavarani et al. (2018), the maximum flying range of drones is set as 15. Besides, the maximum payload weight limit of drones is 30.

For the uncertain demands, we consider 10 scenarios, assuming that the nominal probability distribution follows a discrete probability distribution. We let each scenario $\xi \in \Xi$ has the same nominal probability $1/|\Xi|$ (Guan et al., 2024). The uncertain demands are generated randomly between 20 and 40. Regarding the ambiguity sets, without the loss of generality, we assume the allowable bias limit of the perturbation vector of box ambiguity set ω can take value from the interval $[0,1]$, and the scaling matrix in the ellipsoidal and polyhedral ambiguity set is respectively set as $A_1 = a^E I$ and $A^2 = a^P I$, where I is an identity matrix. The ambiguity scopes can be adjusted by changing the value settings of ω , a^P , and a^E (Qiu et al., 2014).

4.1.2. Computational results

In this part, we analyse the influence of the confidence level α and weight parameter λ on optimal solutions. In this numerical instance, we let $\omega = 0.01$, $a^P = 0.001$, and $a^E = 0.0005$. Tables 3 to 5 depict the computational results of the optimal solution of different risk preferences under the three ambiguity sets. As we can see from Table 3–Table 5, when confidence level α remains fixed, the optimal objective values under three ambiguity sets increase as weight parameter λ increases. Furthermore, for a fixed value of weight parameter λ , a larger α value will also result in a higher optimal solution. This can be attributed to the fact that α serves as an indicator of risk preferences. Specifically, when the α value approaches 1, the resulting solution is the most conservative one, indicating the most risk-averse attitude. Vice versa, when the α value approaches 0, the decision maker is risk-neutral. Therefore, higher values of confidence level α and weight parameter λ both correspond to a greater risk level leading to more risk-averse policies. Decision makers should select the combination of these two coefficients flexibly based on their risk preferences.

Furthermore, we investigate how these risk parameter combinations affect the inventory decisions under the three ambiguity sets. In this numerical experiment, we set $\omega = 0.01$, $a^P = 0.05$, and $a^E = 0.0005$. Besides, we respectively fix the confidence level $\alpha = 0.7, 0.8, 0.9$ and vary the weight parameter λ from 0.1 to 0.9 in increments of 0.1. The computational results are shown in Figs. 2 to 4. It is obvious that a larger weight parameter λ leads to a higher quantity of inventory prepositioning of relief items. However, the quantity of inventory prepositioning of relief items does not change monotonically as a function of confidence level α . This variation tendency is in accord with the computational results from Noyan (2012). Therefore, the incorporation of the CVaR criterion within DRO models assumes a pivotal role, which contributes significantly to making diversifying inventory prepositioning decisions based on different risk preferences.

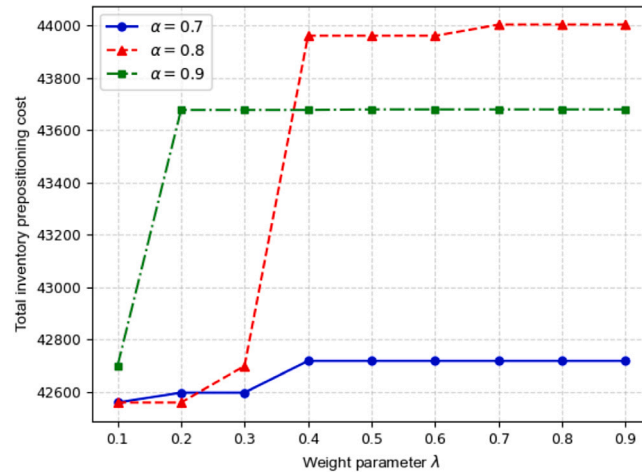


Fig. 2. Total inventory prepositioning cost under the box ambiguity set.

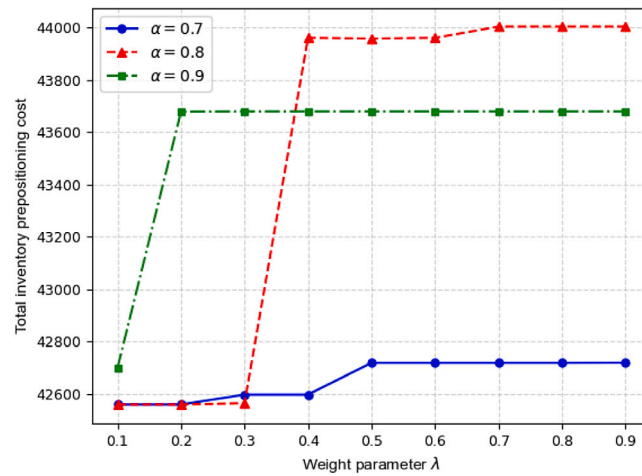


Fig. 3. Total inventory prepositioning cost under the ellipsoidal ambiguity set.

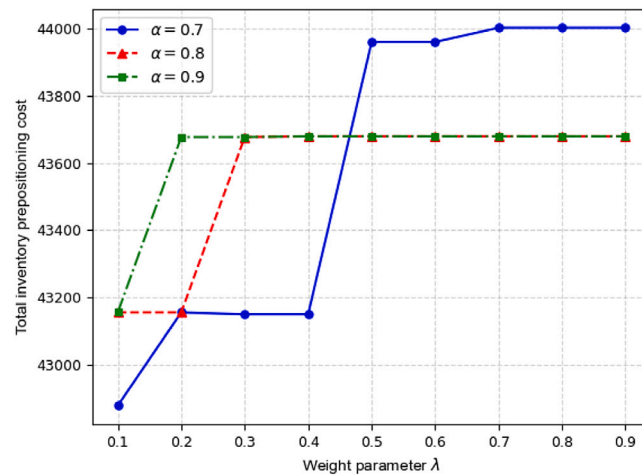


Fig. 4. Total inventory prepositioning cost under the polyhedral ambiguity set.

Table 4

Computational results of optimal objective values under the ellipsoidal ambiguity set.

	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$	$\lambda = 0.9$
$\alpha = 0.5$	54 304.89	54 811.81	55 310.27	55 777.60
$\alpha = 0.6$	54 527.94	55 196.70	55 795.83	56 405.12
$\alpha = 0.7$	54 749.20	55 651.08	56 508.53	57 360.68
$\alpha = 0.8$	55 197.26	56 221.97	57 095.56	57 956.78
$\alpha = 0.9$	55 325.60	56 266.30	57 211.97	58 155.66

Table 5

Computational results of optimal objective values under the polyhedral ambiguity set.

	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$	$\lambda = 0.9$
$\alpha = 0.5$	54 318.23	54 829.00	55 330.84	55 802.42
$\alpha = 0.6$	54 546.04	55 192.12	55 799.60	56 407.07
$\alpha = 0.7$	54 777.44	55 678.00	56 534.97	57 391.94
$\alpha = 0.8$	55 211.56	56 231.12	57 102.04	57 954.47
$\alpha = 0.9$	55 321.98	56 277.93	57 217.22	58 156.51

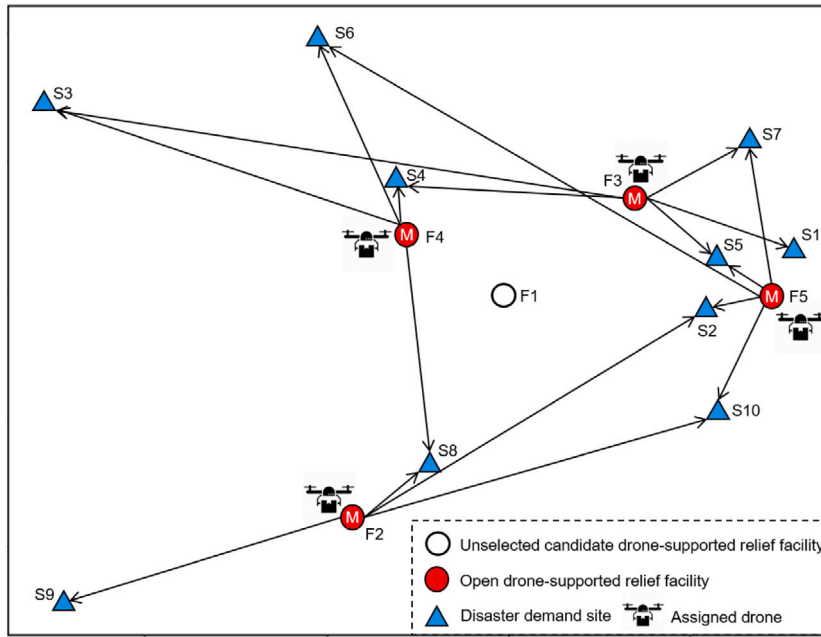
**Fig. 5.** An illustrative example of drone-supported last-mile humanitarian logistics system planning.

Fig. 5 shows the computational results of the drone-supported last-mile humanitarian logistics system planning under the box ambiguity set with $\alpha = \lambda = 0.9$. More specifically, F2 to F5 are designated as open drone-supported relief facilities with medium capacity level, each equipped with one drone. The connections between facilities and disaster demand sites represent the drone visit schedule. For instance, at the drone-supported relief facility F4, the drone is allocated to provide relief services to disaster demand sites S3, S4, S6, and S8. In essence, this implies four drone trips for the drone assigned to the F4 facility.

4.1.3. Analysis of solution quality

In this part, we assess the quality of the solution derived from the proposed model [DRO-DSRFLP] by conducting a series of numerical experiments. Firstly, we analyse the optimal values derived from models [DRO-DSRFLP] and [SP-DSRFLP]. In order to obtain the differences, the price of distributional robustness (PDR) is introduced (Guan et al., 2024) as

$$PDR = \frac{(DRO)^* - (NSO)^*}{(NSO)^*}, \quad (41)$$

where $(NSO)^*$ and $(DRO)^*$ denote the optimal solutions of the stochastic programming model [SP-DSRFLP] and DRO model [DRO-DSRFLP], respectively. Table 6 demonstrates the PDR values under $\alpha = \lambda = 0.9$ with the different adjustable ambiguity scopes. As is illustrated in Table 6, a larger ambiguity scope leads to a higher PDR value. The larger the scope of the ambiguity set, the more probability information is incorporated and considered, leading to a larger additional cost. As the adjustable ambiguity scope

Table 6
Computational results of the PDR.

Box ambiguity set		Ellipsoidal ambiguity set		Polyhedral ambiguity set	
ω	PDR	a^E	PDR	a^P	PDR
0.001	0.0045%	0.00005	0.0002%	0.0005	0.0023%
0.01	0.0454%	0.0001	0.0018%	0.005	0.0227%
0.01	0.4539%	0.0005	0.0025%	0.05	0.227%
0.15	0.5565%	0.001	0.0054%	0.01	0.4539%
0.2	0.644%	0.01	0.0097%	0.5	0.8119%

Table 7
Statistics of out-of-sample optimal total costs derived from model [DRO-DSRFLP] and [SP-DSRFLP].

Model	Mean	25th percentile	75th percentile	95th percentile	Worst
[SP-DSRFLP]	52 622.96	50 519.98	55 159.818	60 536.56	82 869.29
[DRO-DSRFLP] (Box, $\omega = 0.001$)	52 606.14	50 460.12	55 075.23	59 993.73	82 215.01
[DRO-DSRFLP] (Box, $\omega = 0.01$)	52 526.74	50 472.81	55 106.99	60 034.07	79 436.48
[DRO-DSRFLP] (Ellipsoidal, $a^E=0.0001$)	52 434.78	50 393.77	54 883.17	59 665.18	81 456.58
[DRO-DSRFLP] (Ellipsoidal, $a^E=0.001$)	52 522.82	50 475.49	54 982.67	60 257.04	79 320.5
[DRO-DSRFLP] (Polyhedral, $a^P=0.002$)	52 548.61	50 435.30	55 056.18	60 346.68	81 775.27
[DRO-DSRFLP] (Polyhedral, $a^P=0.02$)	52 389.71	50 377.72	54 918.9	59 782.53	80 058.29

decreases, the ambiguity set converges toward a precise probability distribution, leading to the proposed DRO model [DRO-DSRFLP] degenerating into the stochastic programming model [SP-DSRFLP]. Moreover, the maximum PDR value remains below 0.8%. In essence, the proposed DRO model adeptly mitigates the uncertainty in probability distribution at a minimal cost and exhibits better resilience performance when faced with partial distribution information.

Furthermore, we evaluate the solution quality derived from the proposed DRO model [DRO-DSRFLP] by conducting an out-of-sample simulation experiment. More specifically, the stochastic programming model [SP-DSRFLP] and DRO model [DRO-DSRFLP] are solved under the given in-sample data, and the optimal first stage solutions $\{x, y, z, v\}$ are derived respectively. Then we randomly generate 5000 testing samples based on the distribution that we use for generating the in-sample data points. For the decision $\{x, y, z, v\}$ and each test sample, the second-stage recourse problem is solved and the optimal total cost (including the costs incurred in the first stage and second stage) can be calculated. Finally, we obtain 5000 random total costs of model [SP-DSRFLP] and [DRO-DSRFLP], and the solution quality can be evaluated.

Table 7 illustrates the various statistics of optimal total costs derived from model [DRO-DSRFLP] and [SP-DSRFLP] by conducting the out-of-sample experiments. As is shown in Table 7, the model [DRO-DSRFLP] under three ambiguity sets performs better than the [SP-DSRFLP] in terms of total costs on average and at all percentile in both cases. Overall, the findings indicate the proposed model [DRO-DSRFLP] is more adept at mitigating the potential risks caused by ambiguous or partial information about the uncertain parameter. This makes it a more attractive option for humanitarian organisers who are risk-averse. Hence, the better performance of the proposed model [DRO-DSRFLP] highlights the importance of considering the ambiguity of probability distributions related to uncertain parameters in the modelling process.

4.2. A large-scale case study

This subsection implements a case study to verify the proposed models. This study focuses on drone-supported last-mile humanitarian logistics planning of Lushan County located in Sichuan Province, China.

4.2.1. Case background and data description

Lushan County, situated along the Longmenshan fault zone, experienced a seismic event of magnitude 7.0 on April 20, 2013, with a focal depth of approximately 13 km. Subsequent to the earthquake, the aftermath, reported as of 6:00 on April 23, 2013, resulted in 193 casualties, affecting 1.99 million individuals, including tourists. Additionally, 12,211 people were injured, and the total economic loss was estimated at around 85.17 billion. Following the earthquake, rapid responses from local authorities and charitable organisations addressed immediate needs. However, the rolling rocks impeded the path of rescue and transport vehicles, affecting the rescue operations. The Lushan earthquake relief case underscores the crucial importance of incorporating drones into the final stretch of humanitarian logistics, guaranteeing the prompt delivery of emergency relief items.

This case study was originated by Xu et al. (2016), which consists of 5 candidate sites for the distribution centres and 25 disaster points. In this study, we extend the case study with 13 candidate drone-supported relief facilities and 40 disaster sites S1–S40, as is

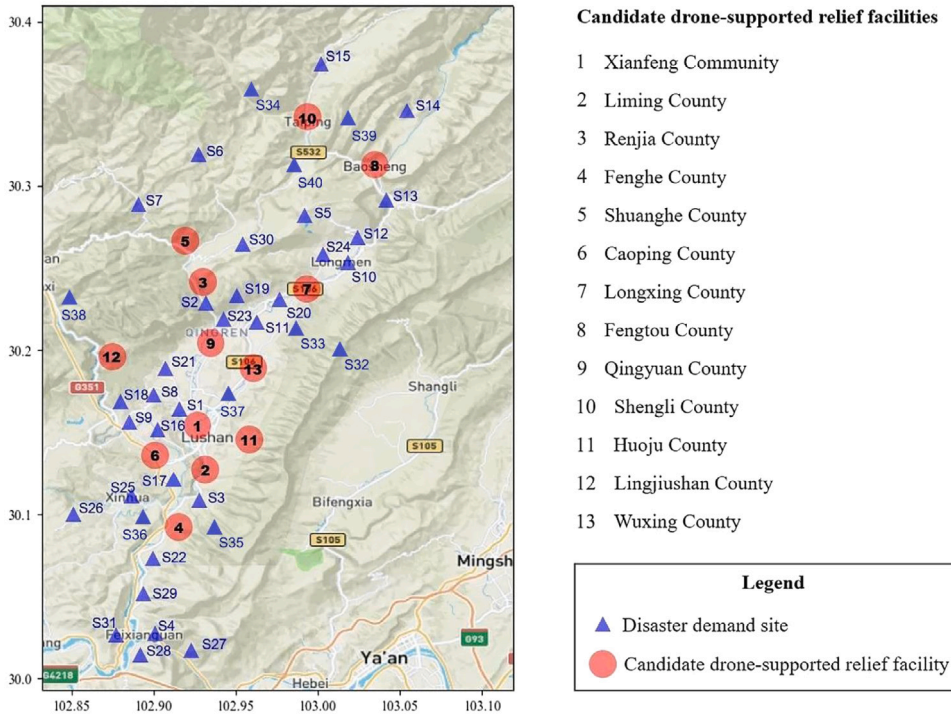


Fig. 6. Case study network.

Table 8
Problem sizes of different scale instances for the case study.

	Small-size	Medium-size	Large-size
Disaster demand sites	15	31	40
Candidate drone-supported relief facilities	13	13	13
Drones	20	30	40
Scenarios	50	200	200
Variables (Box)	210615	2520935	4296975
Constraints (Box)	210794	2520970	4297159
Variables (Ellipsoidal)	210617	2520937	4296977
Constraints (Ellipsoidal)	210796	2520972	4297159
Variables (Polyhedral)	210717	2521337	4297377
Constraints (Polyhedral)	210796	2521370	4297559

shown in Fig. 6. For comparative analysis, we investigate three case study instances with varying problem sizes, i.e., small, medium, and large. For both three case studies, we consider 13 potential drone-supported relief facilities. In the small-size case study, there are 15 disaster demand sites, while there are 31 in the medium-size case study and 40 in the large-size case study. The problem sizes of different scale instances for the case study are reported in Table 8. Moreover, the demands of disaster sites lie within the range of [20,40], with undisclosed occurrence probabilities. We let the nominal probability of each scenario be equal to $1/|\Xi|$. Particularly, for the medium-size and large-size case studies, we designated the inventory capacities of drone-supported relief facilities with small, medium, and large capacity levels as 100, 150, and 250, with opening costs of 500, 550, and 650, respectively. This selection is made to ensure alignment with the requirements of large-scale disaster scenarios, aiming to prevent severe shortages caused by insufficient inventory. As for the other parameters in the medium-size and large-size case studies, the settings are the same as the small-scale example in the last subsection.

4.2.2. Computational results and analysis of small-size case study

In this part, we conducted experiments on a small-scale case study to validate the accuracy of our model. Firstly, we examine the influence of various combinations of confidence level α and weight parameter λ on the proposed model. The computational results under three ambiguity sets are reported in Tables 9 to 11. As can be observed from Tables 9 to 11, the objective value increases when the confidence level α and weight parameter λ become larger under three ambiguity sets. Additionally, the parameters have an influence on the facility locations and the inventory prepositioning. Table 12 lists the results of the inventory prepositioning quantity under three ambiguity sets when confidence level α and weight parameter λ take different values. From Table 12, it can be observed that a larger α or λ will cause a more conservative solution with more quantities of inventory prepositioning.

Table 9

Computational results of the small-size case study under the box ambiguity set.

α	λ	Objective value	Location (Capacity level, Inventory quantity)	CPU time/s	Gap
0.5	0.5	80 641.27	1(S,50); 2(S,50); 4(M,68);5(S,37); 6(S,50); 7(S,50);8(M,73.5);9(S,50); 10(S,35)	214.83	–
	0.7	80 880.85	1(S,50); 2(S,50); 4(S,50);5(S,39); 6(S,50); 7(M,76.73);8(S,50);9(S,50); 10(S,49)	215.94	–
	0.9	81 097.02	1(S,50); 2(S,50); 4(S,50);5(S,40); 6(S,50); 7(M,76.1);8(S,50);9(S,50); 10(S,49)	245.61	–
0.7	0.5	81 322.14	1(S,50); 2(S,50); 4(M,73);5(S,37); 6(S,50); 7(S,50);8(M,71.54);9(S,50); 10(S,35)	205.49	–
	0.7	81 728.32	1(S,50); 2(S,50); 4(M,76.68);5(S,37); 6(S,50); 7(S,50);8(M,69);9(S,50); 10(S,35)	236.42	–
	0.9	82 127.71	1(S,50); 2(S,50); 4(M,77);5(S,38); 6(S,50); 7(S,50);8(M,69.3);9(S,50); 10(S,35)	210.36	–
0.9	0.5	82 019.01	1(S,50); 2(S,50); 4(S,50);5(S,41); 6(S,50); 7(S,50);8(M,71.69);9(S,37); 10(S,35); 13(S,34)	225.36	–
	0.7	82 657.30	1(S,48.12); 2(S,50); 4(M,68);5(S,35); 6(S,50); 7(S,50);8(M,70);9(S,37); 10(S,32); 13(S,29)	216.16	–
	0.9	83 267.32	1(S,50); 2(S,50); 4(M,74);5(S,31); 6(S,50); 7(S,50);8(S,50);9(S,40.5); 10(S,44.15); 13(S,30)	181.3	–

Table 10

Computational results of the small-size case study under the ellipsoidal ambiguity set.

α	λ	Objective value	Location (Capacity level, Inventory quantity)	CPU time/s	Gap
0.5	0.5	80 289.43	1(S,48.72); 2(S,50); 4(S,50);5(S,40); 6(S,50); 7(M,73.27);8(S,50);9(S,50); 10(S,45)	7200	0.39%
	0.7	80 607.03	1(S,50); 2(S,50); 4(M,68.56);5(S,37); 6(S,50); 7(S,70);8(M,73);9(S,50); 10(S,35)	7200	0.11%
	0.9	81 074.80	1(S,50); 2(S,50); 4(M,77.33);5(S,40); 6(S,50); 7(S,50);8(S,50);9(S,50); 10(S,48)	7200	0.13%
0.7	0.5	80 835.59	1(S,50); 2(S,50); 4(M,71.43);5(S,37); 6(S,50); 7(S,50);8(M,73);9(S,50); 10(S,35)	7200	0.03%
	0.7	81 436.53	1(S,50); 2(S,50); 4(M,78);5(S,36.67); 6(S,50); 7(S,50);8(M,68);9(S,50); 10(S,35)	7200	0.01%
	0.9	82 100.03	1(S,50); 2(S,50); 4(S,50);5(S,40); 6(S,50); 7(M,75);8(M,67);9(S,50); 10(S,37)	7200	0.09%
0.9	0.5	81 572.43	1(S,50); 2(S,50); 4(S,50);5(S,40); 6(S,50); 7(M,74.16);8(M,69);9(S,50); 10(S,35)	7200	0.03%
	0.7	82 418.59	1(S,48.11); 2(S,49.8); 4(S,50);5(S,41); 6(S,50); 7(S,50);8(M,71.07);9(S,37); 10(S,35); 13(S,36.82)	7200	0.05%
	0.9	83 197.07	1(S,50); 2(S,50); 4(M,74);5(S,30); 6(S,50); 7(S,50);8(S,50);9(S,41.44); 10(S,45); 13(S,29.23)	7200	0.03%

Furthermore, we conduct extensive sensitive analysis with respect to some parameters to evaluate the impact on the model. Firstly, we explore the impacts of the shortage penalty coefficient on the penalty cost of unsatisfied demand and unused relief items. Figs. 7 to 9 depict the computational results of the penalty costs when the shortage penalty coefficient takes different values. As is shown in Figs. 7 to 9, it is evident that the penalty cost for unsatisfied demand decreases as the shortage penalty coefficient becomes smaller. Conversely, the penalty cost for unused inventory increases with the rise of the shortage penalty coefficient. The findings emphasise the importance of selecting an appropriate shortage penalty coefficient during the different stages of disaster relief. Careful consideration of this coefficient can decrease unsatisfied demand, contributing to more effective relief operations and greatly reducing casualties. Decision makers should conduct a thorough evaluation of the prevailing conditions and context to make informed decisions regarding the shortage penalty coefficient. Moreover, recognising the varying conditions across different disaster districts, the shortage penalty coefficient can also be reasonably adjusted.

In addition, due to the unpredictable nature of meteorological factors and potential progress in battery technology, the maximum flying range of drones can be treated as a variable parameter. Therefore, numerical experiments are conducted to examine the influence caused by the parameter. In humanitarian logistics, an egalitarian policy is crucial and can be measured by the total delivery time to beneficiaries (Duran et al., 2011). To investigate the impact of the maximum flying range of drones on the total delivery time, we conducted a further sensitivity analysis, obtaining the total delivery time under different maximum flying ranges. The resulting variations in the total delivery time under three ambiguity sets are shown in Fig. 10. It can be concluded from Fig. 10 that the total delivery time exhibits significant fluctuation and peaks when the maximum flying range is set to 25. The

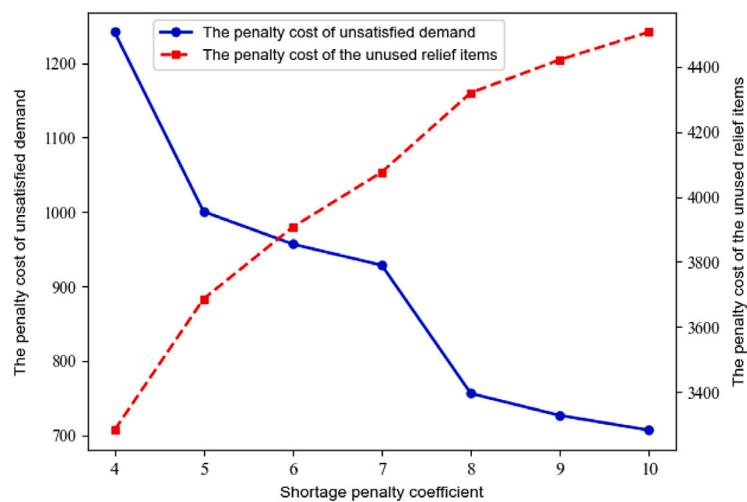


Fig. 7. The impacts of the shortage penalty coefficients on unsatisfied demand and unused relief items under the box ambiguity set.

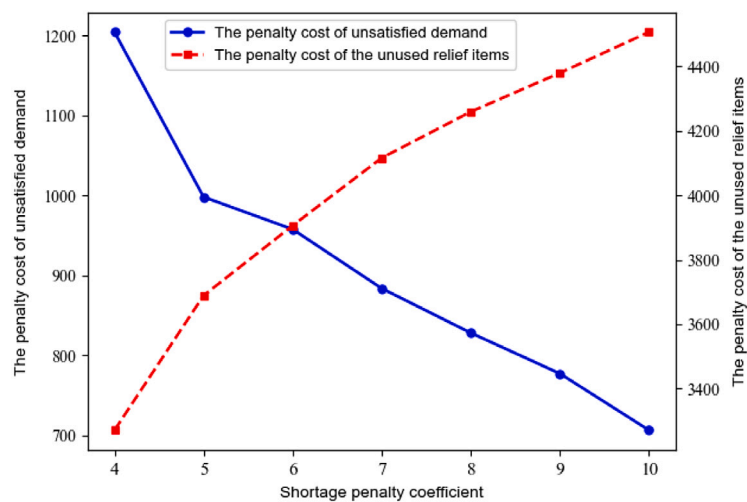


Fig. 8. The impacts of the shortage penalty coefficients on unsatisfied demand and unused relief items under the ellipsoidal ambiguity set.

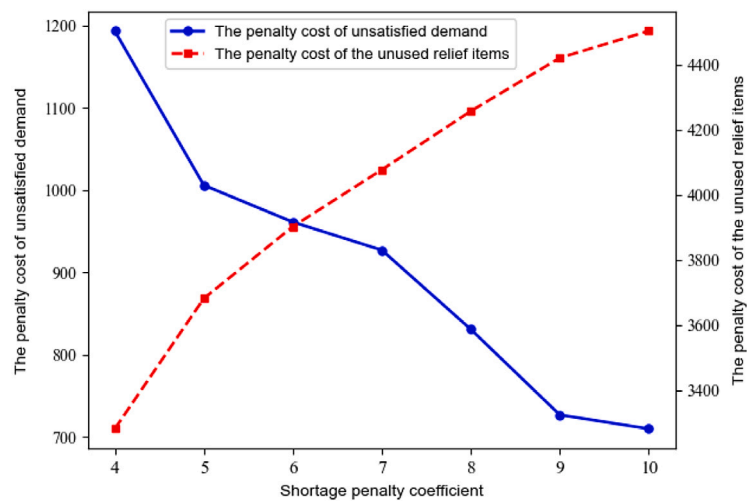


Fig. 9. The impacts of the shortage penalty coefficients on unsatisfied demand and unused relief items under the polyhedral ambiguity set.

Table 11
Computational results of the small-size case study under the polyhedral ambiguity set.

α	λ	Objective value	Location (Capacity level, Inventory quantity)	CPU time/s	Gap
0.5	0.5	80 171.50	1(S,50); 2(S,50); 4(M,68);5(S,37); 6(S,50); 7(S,50);8(M,73.5);9(S,50); 10(S,35)	202.11	–
	0.7	80 655.29	1(S,50); 2(S,50); 4(S,50);5(S,39); 6(S,50); 7(M,76.73);8(S,50);9(S,50); 10(S,49)	193.93	–
	0.9	81 102.39	1(S,50); 2(S,50); 4(S,50);5(S,40); 6(S,50); 7(M,76.1);8(S,50);9(S,50); 10(S,49)	182.27	–
0.7	0.5	80 913.03	1(S,50); 2(S,50); 4(M,73);5(S,37); 6(S,50); 7(S,50);8(M,71.54);9(S,50); 10(S,35)	225.88	–
	0.7	81 528.58	1(S,50); 2(S,50); 4(M,76.68);5(S,37); 6(S,50); 7(S,50);8(M,69)9(S,50); 10(S,35)	161.07	–
	0.9	82 102.27	1(S,50); 2(S,50); 4(M,77);5(S,38); 6(S,50); 7(S,50);8(M,69.3);9(S,50); 10(S,35)	163.18	–
0.9	0.5	81 615.63	1(S,50); 2(S,50); 4(S,50);5(S,41); 6(S,50); 7(S,50);8(M,71.69);9(S,37); 10(S,35); 13(S,34)	285.05	–
	0.7	82 425.90	1(S,48.12); 2(S,50); 4(M,68);5(S,35); 6(S,50); 7(S,50);8(M,70);9(S,37); 10(S,32); 13(S,29)	210.87	–
	0.9	83 209.17	1(S,50); 2(S,50); 4(M,74);5(S,31); 6(S,50); 7(S,50);8(S,50);9(S,40.5); 10(S,44.15); 13(S,30)	277.76	–

Table 12
Computational results the quantity of the inventory prepositioning under three ambiguity sets.

α	λ	Total inventory (Box)	Total inventory (Ellipsoidal)	Total inventory (Polyhedral)
0.5	0.5	463.5	456.99	463.03
	0.7	464.73	463.56	463.48
	0.9	465.1	465.33	464.69
0.7	0.5	466.54	466.42	466
	0.7	467.68	467.67	467.32
	0.9	469.3	469	469.3
0.9	0.5	468.69	468.16	468.67
	0.7	469.12	468.72	468.75
	0.9	469.65	469.67	469.61

fluctuation can be attributed to the distinct configurations of last-mile humanitarian logistics networks resulting from variations in the maximum flying range of drones. Therefore, decision makers should carefully consider the maximum flying range as a crucial parameter when selecting drones for disaster relief operations, incorporating the egalitarian objective into drone-supported last-mile humanitarian logistics planning. Furthermore, tailoring drone specifications based on the specific requirements of the relief mission and the geographical characteristics of the affected areas is essential. Finally, it is also necessary to implement real-time monitoring systems for drone battery performance.

As a final sensitivity analysis, the maximum payload weight limit of drones is varied from 10 to 30 in increments of 5. The computational results of the total cost under three ambiguity sets are shown in Fig. 11. From Fig. 11, it can be found that a larger maximum payload weight limit of drones leads to a significant decrease in total cost. Increasing the maximum payload weight limit can lead to a more efficient use of drone resources and, subsequently, a reduction in the total cost of operations. Drones with larger payload capacities can facilitate the delivery of more relief supplies in a single trip, thereby enhancing the overall delivery capabilities. Therefore, from the managerial perspective, decision makers should explore opportunities to invest in drones with higher payload capacities to achieve cost savings in disaster relief missions, especially in situations requiring large-scale and rapid relief efforts.

4.2.3. Computational results of medium-size and large-size case studies

In this part, a series of numerical experiments are conducted on both medium-size and large-size case studies to further validate the effectiveness of the DRO model. We set the computational time limit to 3 h. The computational results of the medium-size and large-size case studies for various combinations of the confidence level α and weight parameter λ under three ambiguity sets are presented in Table 13 and Table 14, respectively.

From Table 13, it can be observed that the computational gap obtained by Gurobi is below 5% for most case instances and below 6.5% for all case instances in the medium-size case study, indicating the acceptability of the solutions. Additionally, the inventory prepositioning quantities under three ambiguity sets of the medium-size case study are illustrated in Fig. 12. From this figure, we observe that the risk parameter combinations significantly influence the total inventory quantity. Furthermore, it can be observed from Table 14 that the computational gap obtained by Gurobi is below 10% for all case instances, further demonstrating the computational feasibility of the proposed model. This indicates that the model's performance in terms of computational

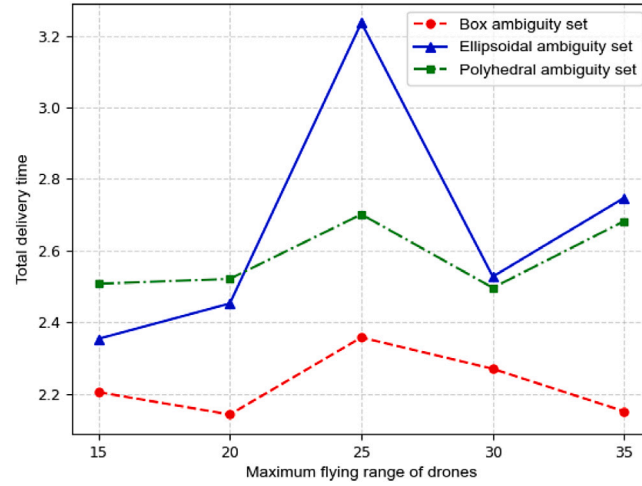


Fig. 10. The impacts of different maximum flying distances of drones on the total delivery time.

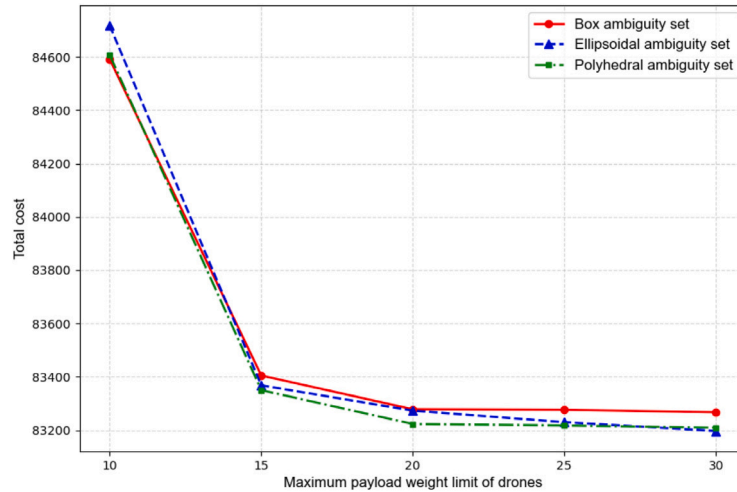


Fig. 11. The impacts of different maximum payload weight limits of drones on the total cost.

Table 13

Computational results of the medium-size case study under three ambiguity sets.

Ambiguity set	α	λ	Objective value	Location (Capacity level)	Gap
Box	0.5	0.7	177 976.72	1(S); 2(L); 3(S); 4(M); 5(S); 6(M); 7(M); 10(S); 12(S); 13(S)	5.34%
	0.9	0.1	174 756.01	1(S); 2(L); 3(S); 4(S); 5(S); 6(L); 7(L); 8(S); 10(S); 12(S)	3.94%
	0.9	0.9	180 001.19	2(L); 4(S); 6(L); 7(M); 8(S); 11(L)	1.17%
Ellipsoidal	0.5	0.7	179 464.94	2(L); 6(L); 7(M); 10(S); 11(S)	6.24%
	0.9	0.1	172 888.30	2(L); 6(L); 7(M); 8(S); 10(S); 11(S)	3.54%
	0.9	0.9	183 611.56	2(L); 4(M); 6(L); 7(M); 8(S); 11(M)	3.54%
Polyhedral	0.5	0.7	173 993.75	2(L); 3(S); 4(S); 6(L); 7(L); 8(S); 9(S); 10(S); 12(S); 13(S)	2.46%
	0.9	0.1	178 559.80	2(L); 6(L); 7(L); 11(L)	6.13%
	0.9	0.9	179 874.83	2(L); 3(S); 4(S); 5(S); 6(L); 7(L); 8(S); 10(S); 12(S)	0.89%

efficiency remains satisfactory across different problem sizes, enhancing its reliability and applicability in real-world decision-making processes.

4.3. Managerial implications and insights

The above numerical experiments can provide several significant managerial implications and insights for humanitarian organisers. The following summary highlights our key findings.

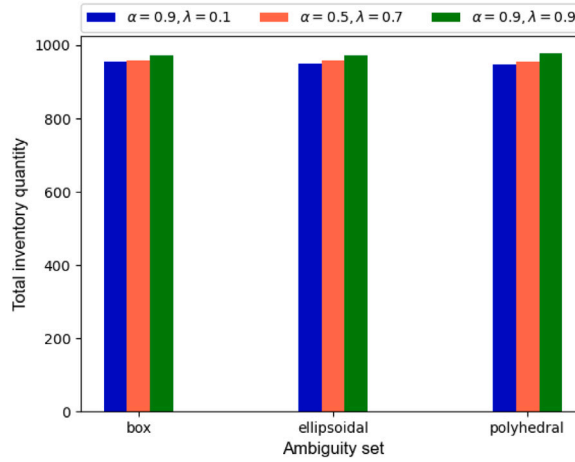


Fig. 12. Total inventory quantity of the large-scale case study under the three ambiguity sets.

Table 14

Computational results of the large-size case study under three ambiguity sets.

Ambiguity set	α	λ	Objective value	Location (Capacity level)	Gap
Box	0.5	0.7	220158.04	2(L); 6(L); 7(L); 10(L); 11(L)	3.28%
	0.9	0.1	219598.15	2(L); 6(L); 7(M); 8(M); 10(M); 11(L)	3.65%
	0.9	0.9	225426.93	1(S); 2(L); 3(S); 4(S); 5(S); 6(L); 7(M); 8(S); 10(M); 12(M); 13(M)	0.86%
Ellipsoidal	0.5	0.7	223367.03	2(L); 4(M); 5(S); 6(L); 7(M); 10(M); 11(L)	5.05%
	0.9	0.1	218410.17	2(L); 6(L); 7(L); 10(L); 11(L)	3.66%
	0.9	0.9	237566	2(L); 4(S); 6(L); 7(M); 8(S); 10(M); 11(L); 13(S)	6.15%
Polyhedral	0.5	0.7	225707.98	2(L); 6(L); 7(L); 10(L); 11(L)	5.13%
	0.9	0.1	215817.87	2(L); 4(S); 6(L); 7(M); 8(S); 10(M); 11(L)	1.85%
	0.9	0.9	244675.91	1(S); 2(L); 4(S); 5(S); 6(L); 7(L); 8(S); 10(L); 12(L); 13(S)	8.54%

(1) Firstly, our study focuses on drone-supported last-mile humanitarian operations in data-scarce disaster scenarios. Computational results demonstrate the better performance of the proposed DRO model in handling uncertain demand with ambiguous distribution. Despite additional costs, the DRO model mitigates the unrealistic assumption of decision makers having complete distribution knowledge, effectively addressing distributional ambiguity in uncertain demand. In terms of managerial insights, humanitarian organisers can strategically employ the DRO model to mitigate the influence of ambiguous demand information in decision-making during unexpected disasters.

(2) The different risk parameter combinations impact the optimal objective and decision outcomes. Larger values for the risk parameters indicate a higher degree of risk aversion by the decision makers. Consequently, humanitarian organisers should judiciously select these parameters based on risk preferences and prevailing disaster circumstances.

(3) Decision makers should select an appropriate shortage penalty coefficient at different stages of disaster relief. Underestimating the coefficient could result in excess unsatisfied demand, leading to substantial disaster losses, while overestimation may yield unnecessary unused inventory. A deliberate consideration of this coefficient can minimise unsatisfied demand and achieve effective inventory management, which helps enhance the efficacy of relief operations and significantly reduce casualties. Humanitarian organisers ought to thoroughly assess prevailing conditions and context to make well-informed decisions about the shortage penalty coefficient, with the flexibility to adjust it based on varying conditions across different disaster districts.

(4) Finally, regarding drone fleet management, drones with larger payload capacities enable the delivery of more relief supplies in a single trip, thereby improving overall delivery capabilities. Increasing the maximum payload weight limit enhances the efficient use of drone resources, leading to a reduction in overall operational costs. Consequently, decision makers should consider investing in drones with higher payload capacities to realise cost savings, especially in situations requiring large-scale and swift relief efforts. Additionally, the maximum flying range of drones has a significant influence on the egalitarian objective indicated by total delivery time. Therefore, tailoring drone specifications to align with the unique demands of relief missions and the geographical characteristics of the operational area becomes imperative for effective and efficient drone deployment. Furthermore, it is crucial to implement real-time monitoring systems for drone battery performance to ensure the reliable operation of drones throughout their missions.

5. Model extensions

We now explore extensions of the fundamental model aimed at considering multiple relief items and incorporating equity constraints.

5.1. Model considering multiple relief items

In the realm of drone-supported disaster relief operations, the consideration of multiple relief items is paramount (Yang et al., 2021). Disasters give rise to a diverse range of needs, spanning from medical kits and blood to vaccines and sanitation provisions. Each relief item is pivotal in meeting the varied requirements of affected populations and expediting their recovery process. Therefore, incorporating multiple relief items into our model is essential to tailor logistical strategies more effectively to meet the diverse needs of disaster-stricken areas, ultimately enhancing the efficiency and effectiveness of drone-supported last-mile humanitarian logistics operations.

5.1.1. Problem description

In this extended model, we consider multiple relief items to meet the diverse needs within the disaster-affected areas.

We let B denote the set of relief items and $b \in B$. Now we introduce the new decision variables and parameters in this extended model. In the first stage, for each $k \in K$, $b \in B$, we define the non-negative continuous decision variable v_{kb} as the inventory prepositioning quantity of relief item b in the drone-supported relief facility k . The unit inventory prepositioning cost for relief item b is denoted by a non-negative parameter c_b^{INV} . Additionally, the open drone-supported relief facility with capacity level l has limited inventory capacity, denoted by W_l . In the second stage, we let D_{ib}^ξ denote the random demand of relief item b or the disaster demand site i under scenario ξ . For each $i \in I$, $k \in K$, $b \in B$, and $\xi \in \Xi$, we let the non-negative continuous variable w_{ikb}^ξ express the number of relief items b delivered to disaster demand site i from drone-supported relief facility k under scenario ξ . We remark that the drone can carry multiple relief items to the assigned disaster demand area and return to the drone-supported relief facility for each drone trip. The unit weight of relief item b is represented by a non-negative parameter a_b . For each $i \in I$, $k \in K$, $d \in D$, $b \in B$, and $\xi \in \Xi$, we represent the number of relief items b delivered to disaster demand site i by the drone d that is assigned to drone-supported relief facility k under scenario ξ as the non-negative continuous variable q_{idkb}^ξ . The delivery cost for unit relief item b from the drone-supported relief facility k to the disaster demand site i is represented by the non-negative parameter c_{kib}^{DELI} . Finally, for each $i \in I$, $b \in B$, and $\xi \in \Xi$, we let the non-negative continuous variable u_{ib}^ξ express the quantity of unsatisfied demand of disaster demand site i for the relief item b under scenario ξ . And for each $k \in K$, $b \in B$, and $\xi \in \Xi$, the non-negative continuous variable r_{kb}^ξ introduced, which means the quantity of unused inventory of the relief item b at the drone-supported relief facility k under scenario ξ . The penalty cost for per unit unsatisfied demand of relief item b and unused relief item b are represented by the non-negative parameters c_b^{UNS} and c_b^{UNU} , respectively.

5.1.2. Mathematical programming formulation

Using the additional notation to the basic DSRFLP, the extended risk-averse stochastic model considering multiple relief items is as follows:

$$\min \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} \sum_{b \in B} c_b^{\text{INV}} v_{kb} + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + (1 - \lambda) \mathbb{E}[U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))] + \lambda \text{CVaR}[U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))] \quad (42)$$

$$\text{s.t.} \quad \sum_{b \in B} v_{kb} \leq \sum_{l \in L} W_l x_{kl}, \quad \forall k \in K, \quad (43)$$

$$v_{kb} \geq 0, \quad \forall k \in K, \forall b \in B, \quad (44)$$

Constraints (2)–(5), (7)–(12),

$$\text{where} \quad U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) = \min \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} \sum_{b \in B} c_{kib}^{\text{DELI}} q_{idkb}^\xi + \sum_{i \in I} \sum_{b \in B} c_b^{\text{UNS}} u_{ib}^\xi + \sum_{k \in K} \sum_{b \in B} c_b^{\text{UNU}} r_{kb}^\xi \quad (45)$$

$$\text{s.t.} \quad \sum_{k \in K} w_{ikb}^\xi + u_{ib}^\xi = D_{ib}^\xi, \quad \forall i \in I, \forall b \in B, \quad (46)$$

$$\sum_{i \in I} w_{ikb}^\xi + r_{kb}^\xi = v_{kb}, \quad \forall k \in K, \forall b \in B, \quad (47)$$

$$w_{ikb}^\xi - \sum_{d \in D} q_{idkb}^\xi = 0, \quad \forall k \in K, \forall i \in I, \forall b \in B, \quad (48)$$

$$\sum_{b \in B} a_b q_{idkb}^\xi \leq L^{\text{MAX}} z_{idk}, \quad \forall k \in K, \forall i \in I, \forall d \in D, \quad (49)$$

$$w_{ikb}^\xi \geq 0, \quad \forall k \in K, \forall i \in I, \forall b \in B, \quad (50)$$

$$q_{idkb}^\xi \geq 0, \quad \forall k \in K, \forall i \in I, \forall d \in D, \forall b \in B, \quad (51)$$

$$u_{ib}^\xi \geq 0, \quad \forall i \in I, \forall b \in B, \quad (52)$$

$$r_{kb}^\xi \geq 0, \quad \forall k \in K, \forall b \in B. \quad (53)$$

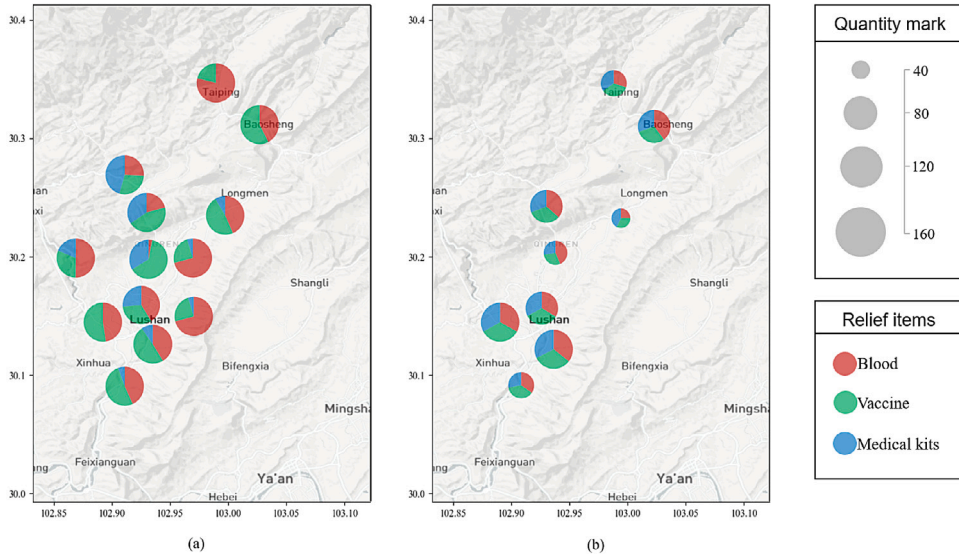


Fig. 13. The inventory of multiple relief items under box ambiguity set — (a) High demand level; (b) Low demand level.

The objective function (42) minimises the cost of the first stage decisions and the expected cost of the second stage decisions. Constraints (43) enforce the inventory restrictions on relief items. Formula (45) defines the new recourse problem considering multiple relief items. Constraints (46) and (47) impose the flow balance of multiple relief items at the disaster area and drone-supported relief facility. Constraints (48) determine the individual delivery quantity of each drone. Constraints (49) require the total number of multiple relief items delivered by the drone not to exceed the maximum payload weight limit. Constraints (44) and (50)–(53) specify the domains of variables.

Similar to the proof logic of Theorems 1–3, we can derive the corresponding tractable DRO models of the extended model under box, ellipsoidal, and polyhedral ambiguity sets. In Appendix F, we provide the specific DRO model formulations.

5.1.3. Computational experiments

We utilise instances from the aforementioned case study with $|I| = 30$, $|J| = 13$, $|D| = 20$, and $|\Xi| = 50$ to validate the extended model. Specifically, we examine three types of relief items: blood, vaccines, and medical kits. The inventory capacities of drone-supported relief facilities with small, medium, and large capacity levels are as follows: 80, 120, and 160. The unit weight of blood, vaccines, and medical kits is 1, 1.2, and 1.5, respectively. Additionally, regarding the parameters related to cost, the holding costs of blood, vaccines, and medical kits are 140, 120, and 100. The unit transportation cost per unit distance for medical kits, vaccines, and blood is 4, 4, and 5. The other parameter settings remain consistent with the above case study. In this experiment, we simulate varying demand levels (high and low) to represent distinct stages of disaster relief. High demand for relief items corresponds to the early stages, while low demand for relief items corresponds to the later stages. We determine the demand range for each relief item across various demand level scenarios as follows: 20 to 40 for the high demand level and 5 to 15 for the low demand level.

The computational results for different demand levels under three ambiguity sets are presented in Figs. 13 to 15. It is evident from these figures that demand levels exert a significant influence on inventory decisions regarding multiple relief items. For instance, in scenarios with high demand levels, nearly all candidate drone-supported relief facilities are operational, and the chosen capacity levels are the largest, resulting in large inventory quantities. Therefore, it is crucial to consider demand variability when making inventory decisions for multiple relief items in drone-supported last-mile humanitarian logistics systems.

Expanding on the managerial implications, these findings highlight the necessity for adaptive inventory management strategies that can accommodate varying demand levels in disaster scenarios. Implementing dynamic inventory policies that adjust facility operations and resource allocation based on real-time demand information can enhance the responsiveness and effectiveness of relief efforts. Additionally, prioritising the establishment of drone-supported relief facilities in areas with anticipated high demand can help ensure the timely and sufficient delivery of essential relief items to affected populations. By proactively addressing demand variability, humanitarian organisations can better allocate resources and mitigate the impact of disasters on vulnerable communities.

5.2. Model considering equity constraint in priority setting

In the realm of drone-supported disaster relief operations, it is crucial to recognise the varying impacts across different disaster-affected regions and the diverse requirements for different types of relief items (Cheng et al., 2021). Therefore, prioritising resource allocation is essential to address the most critical needs promptly, minimising loss and maximising effectiveness. Simultaneously, equity is a significant objective that needs to be considered in humanitarian operations (Farahani et al., 2020; Çankaya et al., 2019).

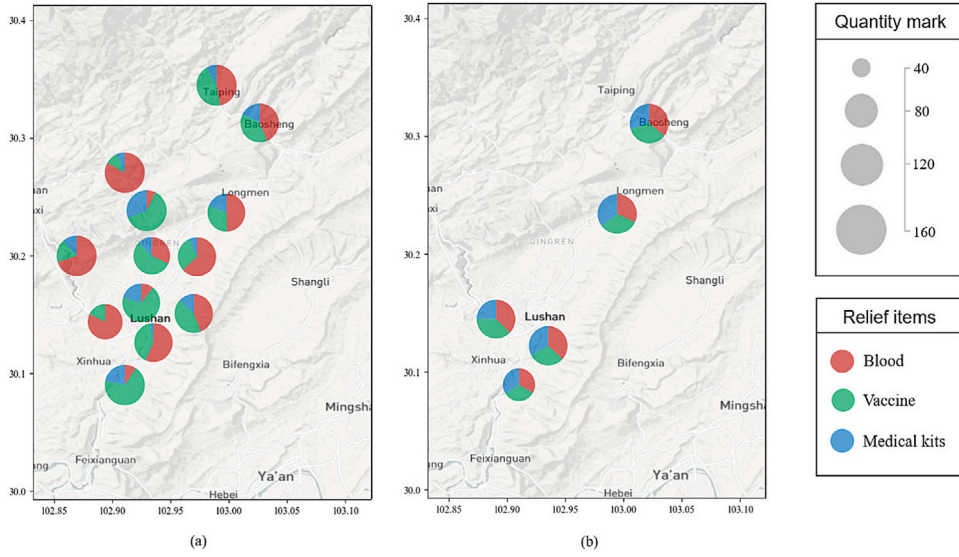


Fig. 14. The inventory of multiple relief items under ellipsoidal ambiguity set — (a) High demand level; (b) Low demand level.

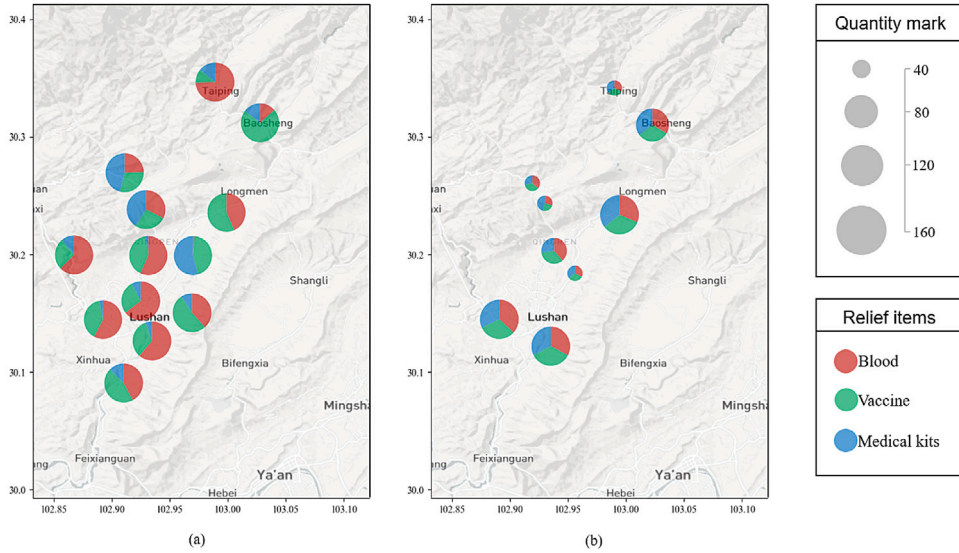


Fig. 15. The inventory of multiple relief items under polyhedral ambiguity set — (a) High demand level; (b) Low demand level.

Disparities resulting from unequal distribution of relief resources can exacerbate existing inequalities and impede humanitarian efforts, particularly in scenarios of severe shortage of relief resources. By incorporating equity constraints into the model, a fair distribution of resources can be guaranteed, considering the specific needs and severity of each disaster-stricken area. This integration not only enhances the efficiency and efficacy of drone-supported last-mile humanitarian logistics but also promotes social justice and welfare. Therefore, we propose an extended model incorporating equity constraints in priority setting to consider priority and equity simultaneously, optimising resource utilisation and fostering social equity amidst crises.

5.2.1. Problem description

In this problem, we integrate the consideration of relief item priority to facilitate the equitable distribution of relief resources.

Specifically, we denote Φ_{ib} as the priority assigned to relief item b at disaster demand site i . This parameter is related to the severity degree of the disaster sites and the importance of the relief item type for each specific site. Note that the greater the priority assigned to a relief item at a disaster site, the higher the penalty cost for unsatisfied demand. Concurrently, we introduce an equity constraint in priority setting to consider the priority and equity simultaneously. We introduce an equity tolerance coefficient, denoted by the non-negative parameter Δ , to ensure a specific level of distributional equity. This coefficient guarantees that the

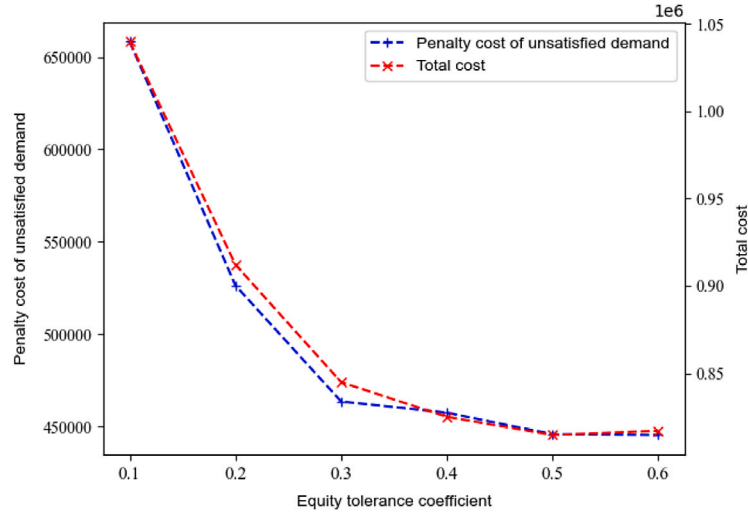


Fig. 16. The impacts of equity tolerance coefficients on total cost and penalty cost of unsatisfied demand under box ambiguity set.

difference in the ratio of unsatisfied demand between two disaster demand sites is within a certain degree. Accordingly, we propose a novel extended model considering equity constraint in priority setting to optimise the reasonable utilisation of relief resources and enhance social welfare simultaneously.

5.2.2. Mathematical programming formulation

Using the additional notation to the above-extended model considering multiple relief items, the extended risk-averse stochastic model considering equity constraint in priority setting can be formulated as follows.

$$\begin{aligned} \min \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} \sum_{b \in B} c_b^{\text{INV}} v_{kb} + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} \\ & + (1 - \lambda) \mathbb{E}[U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))] + \lambda \text{CVaR}[U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))] \end{aligned} \quad (54)$$

s.t. Constraints (2)–(5), (7)–(12), (43)–(44),

$$\text{where } U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) = \min \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} \sum_{b \in B} c_{kib}^{\text{DELI}} q_{idkb}^{\xi} + \sum_{i \in I} \sum_{b \in B} c_b^{\text{UNS}} \Phi_{ib} u_{ib}^{\xi} + \sum_{k \in K} \sum_{b \in B} c_b^{\text{UNU}} r_{kb}^{\xi} \quad (55)$$

$$\text{s.t. } -\Delta \leq \frac{\sum_{b \in B} u_{ib}^{\xi}}{\sum_{b \in B} D_{ib}^{\xi}} - \frac{\sum_{b \in B} u_{jb}^{\xi}}{\sum_{b \in B} D_{jb}^{\xi}} \leq \Delta, \quad \forall i, j \in I, i \neq j, \forall \xi \in \Xi, \quad (56)$$

Constraints (46)–(53).

The objective function (54) minimises the cost of the first stage decisions and the expected cost of the second stage decisions. Formula (55) defines the new recourse problem considering relief item priority. Constraints (56) enforce that the difference in the ratio of the quantity of unsatisfied demand to the total demand between any two disaster demand sites cannot exceed the equity tolerance coefficient, ensuring a certain degree of equity.

Similar to the proof logic of Theorems 1–3, we can derive the corresponding tractable DRO models of the extended model under box, ellipsoidal, and polyhedral ambiguity sets. In Appendix G, we provide the specific DRO model formulations.

5.2.3. Computational experiments

We evaluate the extended model using the same network scale and parameter settings as those employed in the previous model extension. The priority of relief items is randomly generated within the range of 1 to 5. We generate the demand for each relief item within the range of [20,40] randomly. In this experiment, we conduct a sensitivity analysis of the parameter equity tolerance coefficient Δ . We vary Δ from 0.1 to 0.6 in increments of 0.1 and examine the total cost and penalty costs of unsatisfied demand under different values of equity tolerance coefficients. Note that the greater the equity tolerance coefficient, the less stringent the equity requirement. The computational results under three ambiguity sets are illustrated from Figs. 16 to 18.

Observing from Figs. 16 to 18, it is evident that both the total cost and the penalty cost of unsatisfied demand decrease with the increase in the equity tolerance coefficient. This implies that stricter equity requirements result in higher costs incurred and a greater occurrence of unsatisfied demand. Additionally, the decreased speed becomes lower as the equity tolerance coefficient increases. From the perspective of disaster relief management, while equity considerations are important in decision-making processes, it is crucial for managers to consider equity requirements. Strict equity requirements may lead to higher unsatisfied demand, potentially compromising operational efficiency. Hence, it is essential to assess the trade-offs between equity and effectiveness when formulating

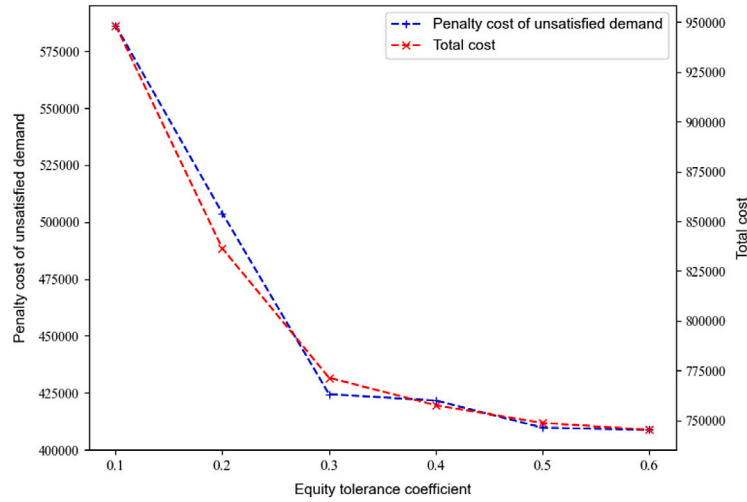


Fig. 17. The impacts of equity tolerance coefficients on total cost and penalty cost of unsatisfied demand under ellipsoidal ambiguity set.

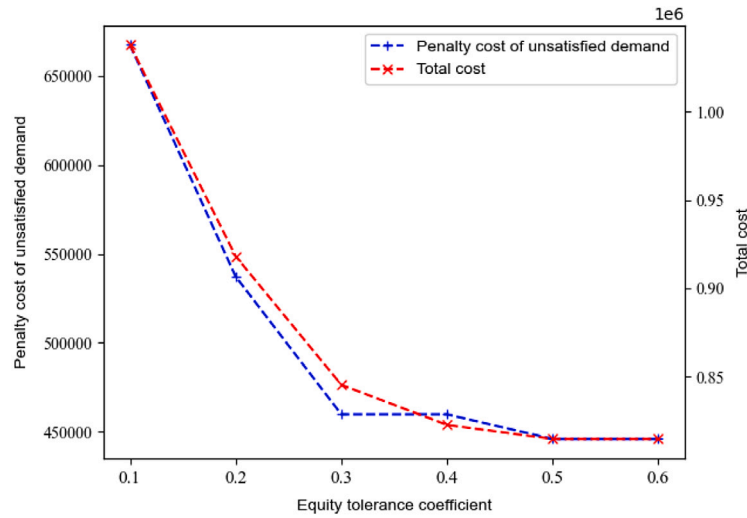


Fig. 18. The impacts of equity tolerance coefficients on total cost and penalty cost of unsatisfied demand under polyhedral ambiguity set.

strategies. In the initial stages of relief operations, the primary focus should be on ensuring the timely delivery of aid to affected areas, even if it means temporarily sacrificing some aspects of equity. This approach helps maximise the effectiveness of relief efforts and saves lives in critical situations. Finally, managers should continuously adjust the equity-related policies as needed. By staying flexible and responsive to changing conditions, organisations can better enhance overall disaster relief performance in the long run.

6. Conclusions and future research

In this paper, we focus on the DSRFLP to make joint decisions considering random demands in the pre-disaster and post-disaster phases to promote the efficient operations of drone-supported last-mile humanitarian logistics. During the pre-disaster phase, our considerations involve determining the drone-supported relief facility location, inventory prepositioning of relief items, drone assignment to the opened facilities, and drone allocation to disaster demand sites. In the post-disaster phase, we address the decisions related to post-disaster delivery quantities in response to various disaster scenarios, which are made after the demand information becomes known. Due to the difficulty of obtaining an accurate probability distribution of the uncertain demands in disaster scenarios, we establish a two-stage DRO model in which uncertain demands can follow uncertain distributions defined by a box, ellipsoidal, and polyhedral ambiguity set to optimise the joint decisions of pre-disaster and post-disaster phases. Additionally, WMCVaR risk measurement is adopted in this DRO model to reflect the risk-averse attitude of humanitarian organisers in chaotic disaster relief systems. To address the challenge of computational intractability, we equivalently reformulate the DRO model with

three ambiguity sets into a computationally tractable one. Furthermore, the validity of the proposed DRO model is verified through a small-scale example and a large-scale case study. The numerical results of the small-scale example demonstrate the proposed DRO model exhibits superior performance in hedging against the influence caused by incomplete information on the uncertain demand. According to the experimental results, we put forth several managerial implications and insights to support the decision-making processes for humanitarian organisations. Finally, we propose two extended models to address multiple relief items and equity constraints in priority setting and perform numerical experiments, aiming to adapt to various real-world disaster scenarios.

This research is the first to apply the DRO method in the DSRFLP to make hybrid decisions in drone-supported last-mile humanitarian logistics planning. However, some limitations can be extended in future research. First, it would be interesting to consider the disruption risk of the drone-supported relief facility. Furthermore, future research will examine how the weather conditions of different disaster areas impact our model framework. Finally, integrating other uncertain factors and exploring alternative ambiguity sets, such as the Wasserstein ambiguity set, to characterise uncertain parameters in this problem would be promising for further investigation.

CRedit authorship contribution statement

Zhongyi Jin: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Kam K.H. Ng:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Chenliang Zhang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation. **Wei Liu:** Writing – review & editing, Visualization, Validation, Resources, Formal analysis. **Fangni Zhang:** Validation, Investigation, Formal analysis, Conceptualization. **Gangyan Xu:** Writing – review & editing, Validation, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Proof of Theorem 1

Proof of Theorem 1. If the probability distribution of the discrete uncertain demand belongs to a box ambiguity set $\mathcal{P}^{\text{Box}}$ that is defined in Eq. (35), the expectation part of the model (34) can be reformulated as

$$\begin{aligned} & \min_{x,y,z,v,\Psi} \max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{p} \\ & = \min_{x,y,z,v,\Psi} \left\{ Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{p}^0 + \max_{\epsilon} [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \epsilon \mid \mathbf{e}^T \epsilon = 0, \|\epsilon\|_{\infty} \leq \omega] \right\}. \end{aligned} \quad (57)$$

Based on the linear programming duality theory, the above model can be reformulated as

$$\min_{x,y,z,v,\Psi} \left\{ Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{p}^0 + \min_{\tau, \zeta, \rho} \left[\boldsymbol{\Omega}^T \zeta + \boldsymbol{\Omega}^T \rho \mid \begin{array}{l} \mathbf{e}^T \tau + \zeta - \rho = Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)), \\ \zeta \geq 0, \rho \geq 0. \end{array} \right] \right\}, \quad (58)$$

where $(\tau, \zeta, \rho) \in \mathbb{R} \times \mathbb{R}_+^{|\Xi|} \times \mathbb{R}_+^{|\Xi|}$ are the dual variables and $\boldsymbol{\Omega} = \omega \mathbf{e}$.

Recall that the WCVaR denotes the worst-case CVaR, and suppose that $\mathcal{P}^{\text{Box}}$ is a compact convex set. Then, the WCVaR $_{\alpha}(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)))$ can be expressed as (Zhu and Fukushima, 2009)

$$\begin{aligned} \text{WCVaR}_{\alpha}(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))) &= \max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \text{CVaR}_{\alpha}(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))) \\ &= \max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \min_{\eta} \eta + \frac{1}{1-\alpha} \sum_{\xi \in \Xi} p^{\xi} [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta]^+ \\ &= \min_{\eta} \max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \eta + \frac{1}{1-\alpha} \sum_{\xi \in \Xi} p^{\xi} [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta]^+. \end{aligned} \quad (59)$$

We can exchange the order of the operators $\max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}}$ and $\min_{\eta}$ mainly because the formula $\eta + \frac{1}{1-\alpha} \sum_{\xi \in \Xi} p^{\xi} [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta]^+$ is convex in η (Rockafellar and Uryasev, 2002) and affine (concave) in $\mathbf{p}$.

Therefore, we can rewrite the WCVaR part in the problem (34) as

$$\min_{x,y,z,v,\Psi} \text{WCVaR}_{\alpha}(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))) = \min_{x,y,z,v,\Psi} \max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \eta + \frac{1}{1-\alpha} \sum_{\xi \in \Xi} p^{\xi} [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta]^+. \quad (60)$$

By introducing an auxiliary vector $\mathbf{s} = \{s_1, s_2, \dots, s_{|\Xi|}\}^T$, we can equivalently rewrite Eq. (60) as the following problem:

$$\min_{x,y,z,v,\Psi,s,\eta} \max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \eta + \frac{1}{1-\alpha} \mathbf{p}^T \mathbf{s} \quad (61)$$

$$\text{s.t. } \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta \mathbf{e}, \quad (62)$$

$$\mathbf{s} \geq \mathbf{0}. \quad (63)$$

Hence, the equivalent formulation of the above problem can be expressed as the following minimisation problem.

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi, \mathbf{s}, \Lambda, \eta} \quad & \Lambda \\ \text{s.t.} \quad & \Lambda \geq \max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \eta + \frac{1}{1-\alpha} \mathbf{p}^\top \mathbf{s}, \end{aligned} \quad (64)$$

Constraints (62)–(63).

In light of the max operation present in the first constraints of problem (64), it becomes imperative to transform this problem into a tractable one. Recall that the distribution probability $\mathbf{p}$ belongs to the box ambiguity set $\mathcal{P}^{\text{Box}}$ expressed in Eq. (35), we have

$$\eta + \frac{1}{1-\alpha} \mathbf{p}^\top \mathbf{s} = \eta + \frac{1}{1-\alpha} (\mathbf{p}^0)^\top \mathbf{s} + \frac{1}{1-\alpha} \epsilon^\top \mathbf{s}. \quad (65)$$

Therefore, the right-side hand of the first constraints of problem (64) can be formulated as

$$\max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \eta + \frac{1}{1-\alpha} \mathbf{p}^\top \mathbf{s} = \eta + \frac{1}{1-\alpha} (\mathbf{p}^0)^\top \mathbf{s} + \frac{Y^*(\mathbf{s})}{1-\alpha}, \quad (66)$$

where $Y^*(\mathbf{s})$ denotes the optimal value of the following linear program.

$$\max_{\epsilon} \{ \epsilon^\top \mathbf{s} \mid \mathbf{e}^\top \epsilon = 0, \|\epsilon\|_\infty \leq \omega \}. \quad (67)$$

Based on the duality theory, the dual program of Eq. (67) is as follows:

$$Y^*(\mathbf{s}) = \min_{\hat{\tau}, \hat{\zeta}, \hat{\rho}} \left\{ \begin{array}{l} \boldsymbol{\Omega}^\top \hat{\zeta} + \boldsymbol{\Omega}^\top \hat{\rho} \mid \\ \mathbf{e}^\top \hat{\tau} + \hat{\zeta} - \hat{\rho} = \mathbf{s}, \\ \hat{\zeta} \geq \mathbf{0}, \quad \hat{\rho} \geq \mathbf{0}. \end{array} \right\}. \quad (68)$$

Hence, we have the following minimisation problem.

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi, \mathbf{s}, \Lambda, \hat{\tau}, \hat{\zeta}, \hat{\rho}, \hat{\tau}, \eta} \quad & \Lambda \\ \text{s.t.} \quad & \Lambda \geq \eta + \frac{1}{1-\alpha} (\mathbf{p}^0)^\top \mathbf{s} + \frac{1}{1-\alpha} (\boldsymbol{\Omega}^\top \hat{\zeta} + \boldsymbol{\Omega}^\top \hat{\rho}), \\ & \mathbf{e}^\top \hat{\tau} + \hat{\zeta} - \hat{\rho} = \mathbf{s}, \\ & \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta \mathbf{e}, \\ & \hat{\zeta} \geq \mathbf{0}, \quad \hat{\rho} \geq \mathbf{0}, \quad \mathbf{s} \geq \mathbf{0}. \end{aligned} \quad (69)$$

Note that solving Problem (64) is equivalent to addressing Problem (69). Proposition 1, proposed in Appendix B, demonstrates the equivalence between these two problems.

Therefore, we can express the formula $\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \text{WMCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)))$ in the problem (32) as

$$\begin{aligned} & \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \text{WMCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))) \\ &= \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} (1-\lambda) \sup_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \mathbb{E}(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))) + \lambda \text{WCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))) \\ &= \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \left\{ \begin{array}{l} \min_{\tau, \zeta, \rho} (1-\lambda) [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^\top \mathbf{p}^0 + \boldsymbol{\Omega}^\top \zeta + \boldsymbol{\Omega}^\top \rho] \mid \\ \quad \mathbf{e}^\top \tau + \zeta - \rho = Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)), \\ \quad \zeta \geq \mathbf{0}, \rho \geq \mathbf{0}. \\ + \lambda \min_{\hat{\tau}, \hat{\zeta}, \hat{\rho}, \hat{\tau}, \eta} \left\{ \begin{array}{l} \eta + \frac{1}{1-\alpha} (\mathbf{p}^0)^\top \mathbf{s} + \frac{1}{1-\alpha} (\boldsymbol{\Omega}^\top \hat{\zeta} + \boldsymbol{\Omega}^\top \hat{\rho}) \mid \\ \quad \mathbf{e}^\top \hat{\tau} + \hat{\zeta} - \hat{\rho} = \mathbf{s}, \\ \quad \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta \mathbf{e}, \\ \quad \hat{\zeta} \geq \mathbf{0}, \quad \hat{\rho} \geq \mathbf{0}, \quad \mathbf{s} \geq \mathbf{0}. \end{array} \right\} \end{array} \right\}. \end{aligned} \quad (70)$$

Finally, the proposed DRO model [DRP-DSRFLP] under the box ambiguity set $\mathcal{P}^{\text{Box}}$ can be equivalently reformulated as follows:

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi, \tau, \zeta, \rho, \hat{\tau}, \hat{\zeta}, \hat{\rho}, \hat{\tau}, \eta} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c_k^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\ & + (1-\lambda) [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^\top \mathbf{p}^0 + \boldsymbol{\Omega}^\top \zeta + \boldsymbol{\Omega}^\top \rho] + \frac{\lambda}{1-\alpha} [(\mathbf{p}^0)^\top \mathbf{s} + \boldsymbol{\Omega}^\top \hat{\zeta} + \boldsymbol{\Omega}^\top \hat{\rho}] \\ \text{s.t.} \quad & \mathbf{e}^\top \tau + \zeta - \rho = Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)), \\ & \mathbf{e}^\top \hat{\tau} + \hat{\zeta} - \hat{\rho} = \mathbf{s}, \\ & \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta \mathbf{e}, \\ & \zeta \geq \mathbf{0}, \rho \geq \mathbf{0}, \hat{\zeta} \geq \mathbf{0}, \hat{\rho} \geq \mathbf{0}, \mathbf{s} \geq \mathbf{0}, \\ & \text{Constraints (2)–(13), Constraints (15)–(22).} \end{aligned} \quad (71)$$

where $(\tau, \zeta, \rho, \hat{\tau}, \hat{\zeta}, \hat{\rho}) \in \mathbb{R} \times \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R} \times \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R}_+^{|\mathcal{I}|}$ are dual variables.

The proof of Theorem 1 is complete. $\square$

Appendix B. Proposition 1

Proposition 1. If $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{z}^*, \mathbf{v}^*, \mathbf{\Psi}^*, \mathbf{s}^*, \Lambda^*, \hat{\xi}^*, \hat{\rho}^*, \hat{\tau}^*, \eta^*)$ solves problem (69), then $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{z}^*, \mathbf{v}^*, \mathbf{\Psi}^*, \mathbf{s}^*, \Lambda^*, \eta^*)$ solves problem (64); Conversely, if $(\tilde{\mathbf{x}}^*, \tilde{\mathbf{y}}^*, \tilde{\mathbf{z}}^*, \tilde{\mathbf{v}}^*, \tilde{\mathbf{\Psi}}^*, \tilde{\mathbf{s}}^*, \tilde{\Lambda}^*, \tilde{\eta}^*)$ solves problem (64), then $(\tilde{\mathbf{x}}^*, \tilde{\mathbf{y}}^*, \tilde{\mathbf{z}}^*, \tilde{\mathbf{v}}^*, \tilde{\mathbf{\Psi}}^*, \tilde{\mathbf{s}}^*, \tilde{\Lambda}^*, \tilde{\xi}^*, \tilde{\rho}^*, \tilde{\tau}^*, \tilde{\eta}^*)$ solves problem (69), where $(\tilde{\xi}^*, \tilde{\rho}^*, \tilde{\tau}^*)$ is the optimal solution of (68).

Proof of Proposition 1. The proof of Proposition 1 is given in Appendix C.

Appendix C. Proof of Proposition 1

We let $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{z}^*, \mathbf{v}^*, \mathbf{\Psi}^*, \mathbf{s}^*, \Lambda^*, \hat{\xi}^*, \hat{\rho}^*, \hat{\tau}^*, \eta^*)$ solves problem (69). Based on the Lagrange weak duality theorem, we have

$$Y^*(\mathbf{s}^*) \leq \Omega^T \hat{\xi}^* + \Omega^T \hat{\rho}^*. \quad (72)$$

Hence, we have

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \eta^* + \frac{1}{1-\alpha} \mathbf{p}^T \mathbf{s}^* &= \eta^* + \frac{1}{1-\alpha} (\mathbf{p}^0)^T \mathbf{s}^* + \frac{Y^*(\mathbf{s}^*)}{1-\alpha} \\ &\leq \eta^* + \frac{1}{1-\alpha} (\mathbf{p}^0)^T \mathbf{s}^* + \frac{1}{1-\alpha} (\Omega^T \hat{\xi}^* + \Omega^T \hat{\rho}^*). \\ &\leq \Lambda^*. \end{aligned} \quad (73)$$

Together with constraints (62) and (63), we can conclude that $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{z}^*, \mathbf{v}^*, \mathbf{\Psi}^*, \mathbf{s}^*, \Lambda^*, \eta^*)$ is a feasible solution to problem (64).

Now we assume that $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{z}^*, \mathbf{v}^*, \mathbf{\Psi}^*, \mathbf{s}^*, \Lambda^*, \eta^*)$ is not an optimal solution to problem (64). There exists an optimal solution $(\underline{\mathbf{x}}^*, \underline{\mathbf{y}}^*, \underline{\mathbf{z}}^*, \underline{\mathbf{v}}^*, \underline{\mathbf{\Psi}}^*, \underline{\mathbf{s}}^*, \underline{\Lambda}^*, \underline{\eta}^*)$ to problem (64) such that

$$\underline{\Lambda}^* < \Lambda^*. \quad (74)$$

We let $(\hat{\xi}^*, \hat{\rho}^*, \hat{\tau}^*)$ be an optimal solution to problem (67). Based on the strong duality theorem, we have

$$\begin{aligned} \underline{\eta}^* + \frac{1}{1-\alpha} (\mathbf{p}^0)^T \underline{\mathbf{s}}^* + \frac{1}{1-\alpha} (\Omega^T \hat{\xi}^* + \Omega^T \hat{\rho}^*) \\ = \underline{\eta}^* + \frac{1}{1-\alpha} (\mathbf{p}^0)^T \underline{\mathbf{s}}^* + \frac{Y^*(\mathbf{s}^*)}{1-\alpha} \\ = \max_{\mathbf{p} \in \mathcal{P}^{\text{Box}}} \eta^* + \frac{1}{1-\alpha} \mathbf{p}^T \underline{\mathbf{s}}^* \leq \underline{\Lambda}^*. \end{aligned} \quad (75)$$

Together with constraints (62), (63), and the constraints in (68), we can find that $(\underline{\mathbf{x}}^*, \underline{\mathbf{y}}^*, \underline{\mathbf{z}}^*, \underline{\mathbf{v}}^*, \underline{\mathbf{\Psi}}^*, \underline{\mathbf{s}}^*, \underline{\Lambda}^*, \hat{\xi}^*, \hat{\rho}^*, \hat{\tau}^*, \eta^*)$ is feasible to problem (69), which contradicts the assumptions that $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{z}^*, \mathbf{v}^*, \mathbf{\Psi}^*, \mathbf{s}^*, \Lambda^*, \hat{\xi}^*, \hat{\rho}^*, \hat{\tau}^*, \eta^*)$ is an optimal solution to problem (69) because $\underline{\Lambda}^* < \Lambda^*$.

Conversely, we let $(\tilde{\mathbf{x}}^*, \tilde{\mathbf{y}}^*, \tilde{\mathbf{z}}^*, \tilde{\mathbf{v}}^*, \tilde{\mathbf{\Psi}}^*, \tilde{\mathbf{s}}^*, \tilde{\Lambda}^*, \tilde{\eta}^*)$ be a solution to (64) and let $(\tilde{\xi}^*, \tilde{\rho}^*, \tilde{\tau}^*)$ represent the optimal solution of (68). It is obvious to find that $(\tilde{\mathbf{x}}^*, \tilde{\mathbf{y}}^*, \tilde{\mathbf{z}}^*, \tilde{\mathbf{v}}^*, \tilde{\mathbf{\Psi}}^*, \tilde{\mathbf{s}}^*, \tilde{\Lambda}^*, \tilde{\xi}^*, \tilde{\rho}^*, \tilde{\tau}^*, \tilde{\eta}^*)$ must be a solution to problem (69). Similarly, we make an assumption that if $(\tilde{\mathbf{x}}^*, \tilde{\mathbf{y}}^*, \tilde{\mathbf{z}}^*, \tilde{\mathbf{v}}^*, \tilde{\mathbf{\Psi}}^*, \tilde{\mathbf{s}}^*, \tilde{\Lambda}^*, \tilde{\xi}^*, \tilde{\rho}^*, \tilde{\tau}^*, \tilde{\eta}^*)$ is not the optimal solution of problem (69), i.e., there exists a solution $(\underline{\mathbf{x}}^*, \underline{\mathbf{y}}^*, \underline{\mathbf{z}}^*, \underline{\mathbf{v}}^*, \underline{\mathbf{\Psi}}^*, \underline{\mathbf{s}}^*, \underline{\Lambda}^*, \hat{\xi}^*, \hat{\rho}^*, \hat{\tau}^*, \eta^*)$ such that $\underline{\Lambda}^* < \tilde{\Lambda}^*$. Following the logic of the first part of this proof, we can find that $(\underline{\mathbf{x}}^*, \underline{\mathbf{y}}^*, \underline{\mathbf{z}}^*, \underline{\mathbf{v}}^*, \underline{\mathbf{\Psi}}^*, \underline{\mathbf{s}}^*, \underline{\Lambda}^*, \eta^*)$ is an optimal solution of problem (64). This contradicts to the assumption that $(\tilde{\mathbf{x}}^*, \tilde{\mathbf{y}}^*, \tilde{\mathbf{z}}^*, \tilde{\mathbf{v}}^*, \tilde{\mathbf{\Psi}}^*, \tilde{\mathbf{s}}^*, \tilde{\Lambda}^*, \tilde{\eta}^*)$ solves problem (64) because $\underline{\Lambda}^* < \tilde{\Lambda}^*$.

The proof of Proposition 1 is complete. $\square$

Appendix D. Proof of Theorem 2

Proof of Theorem 2. If the probability distribution of the discrete uncertain demand belongs to an ellipsoidal ambiguity set $\mathcal{P}^{\text{Ellipsoidal}}$ that is defined in Eq. (36), the expectation part of the model (34) can be reformulated as

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \mathbf{\Psi}} \max_{\mathbf{p} \in \mathcal{P}^{\text{Ellipsoidal}}} Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \mathbf{\Theta}(\xi))^T \mathbf{p} \\ = \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \mathbf{\Psi}} \left\{ Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \mathbf{\Theta}(\xi))^T \mathbf{p}^0 + \max_{\epsilon} [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \mathbf{\Theta}(\xi))^T \mathbf{A}_1 \epsilon \mid \mathbf{e}^T \mathbf{A}_1 \epsilon = 0, \mathbf{p}^0 + \mathbf{A}_1 \epsilon \geq 0, \|\epsilon\|_2 \leq 1] \right\}. \end{aligned} \quad (76)$$

Now we consider the inner maximisation problem of Eq. (76).

$$\max_{\epsilon} [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \mathbf{\Theta}(\xi))^T \mathbf{A}_1 \epsilon \mid \mathbf{e}^T \mathbf{A}_1 \epsilon = 0, \mathbf{p}^0 + \mathbf{A}_1 \epsilon \geq 0, \|\epsilon\|_2 \leq 1]. \quad (77)$$

The above maximisation problem is equivalent to the following minimisation problem.

$$\min_{\epsilon} [-Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{A}_1 \epsilon \mid \mathbf{e}^T \mathbf{A}_1 \epsilon = 0, \mathbf{p}^0 + \mathbf{A}_1 \epsilon \geq 0, \|\epsilon\|_2 \leq 1]. \quad (78)$$

The Lagrange function of the above optimisation problem (78) is as follows:

$$\mathcal{L}(\boldsymbol{\vartheta}, \theta, \zeta; \epsilon) = -Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{A}_1 \epsilon + \boldsymbol{\vartheta}^T (-\mathbf{p}^0 - \mathbf{e}^T \mathbf{A}_1) \epsilon + \theta (\mathbf{e}^T \mathbf{A}_1 \epsilon) + \zeta (\|\epsilon\|_2 - 1), \quad (79)$$

where $(\boldsymbol{\vartheta}, \theta, \zeta) \in \mathbb{R}^{|\Xi|} \times \mathbb{R} \times \mathbb{R}$ are the Lagrange multipliers.

Then the Lagrange dual function can be formulated as

$$\begin{aligned} \mathcal{D}(\boldsymbol{\vartheta}, \theta, \zeta) &= \min_{\epsilon} \mathcal{L}(\boldsymbol{\vartheta}, \theta, \zeta; \epsilon) \\ &= -(\mathbf{p}^0)^T \boldsymbol{\vartheta} - \zeta - \max_{\epsilon} \left\{ [\mathbf{A}_1^T \boldsymbol{\vartheta} + \mathbf{A}_1^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \theta \mathbf{A}_1^T \mathbf{e}]^T \epsilon - \zeta \|\epsilon\|_2 \right\} \\ &= -(\mathbf{p}^0)^T \boldsymbol{\vartheta} - \zeta - \mathcal{R}^* [\mathbf{A}_1^T \boldsymbol{\vartheta} + \mathbf{A}_1^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \theta \mathbf{A}_1^T \mathbf{e}], \end{aligned} \quad (80)$$

where

$$\mathcal{R}^*(\cdot) = \begin{cases} 0 & \|\cdot\|_2 \leq \zeta \\ \infty & \text{otherwise} \end{cases} \quad (81)$$

is the conjugate function of $\mathcal{R} = \zeta \|\epsilon\|_2$ (Boyd and Vandenberghe, 2004). For any $\zeta \geq 0$ and $\boldsymbol{\vartheta} \geq \mathbf{0}$, the Lagrange dual function can yield lower bounds. Hence, we have the following equivalent problem of Eq. (78).

$$\max_{\boldsymbol{\vartheta}, \theta, \zeta} \mathcal{D}(\boldsymbol{\vartheta}, \theta, \zeta) = \max_{\boldsymbol{\vartheta}, \theta, \zeta} \left\{ -(\mathbf{p}^0)^T \boldsymbol{\vartheta} - \zeta \mid \begin{array}{l} \|\mathbf{A}_1^T \boldsymbol{\vartheta} + \mathbf{A}_1^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \theta \mathbf{A}_1^T \mathbf{e}\|_2 \leq \zeta \\ \boldsymbol{\vartheta} \geq \mathbf{0}, \quad \zeta \geq 0. \end{array} \right\}. \quad (82)$$

Hence, we have the equivalent minimisation problem as follows:

$$\min_{\boldsymbol{\vartheta}, \theta, \zeta} \left\{ (\mathbf{p}^0)^T \boldsymbol{\vartheta} + \zeta \mid \begin{array}{l} \|\mathbf{A}_1^T \boldsymbol{\vartheta} + \mathbf{A}_1^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \theta \mathbf{A}_1^T \mathbf{e}\|_2 \leq \zeta \\ \boldsymbol{\vartheta} \geq \mathbf{0}, \quad \zeta \geq 0. \end{array} \right\}. \quad (83)$$

Therefore, we have

$$\begin{aligned} &\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}} \max_{\mathbf{p} \in \mathcal{P}^{\text{Ellipsoidal}}} Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{p} \\ &= \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}, \boldsymbol{\vartheta}, \theta, \zeta} \left\{ Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{p}^0 + (\mathbf{p}^0)^T \boldsymbol{\vartheta} + \zeta \mid \begin{array}{l} \|\mathbf{A}_1^T \boldsymbol{\vartheta} + \mathbf{A}_1^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \theta \mathbf{A}_1^T \mathbf{e}\|_2 \leq \zeta \\ \boldsymbol{\vartheta} \geq \mathbf{0}, \quad \zeta \geq 0. \end{array} \right\}. \end{aligned} \quad (84)$$

Following the logic of the proof of Theorem 1, we can derive the equivalent problem of the $\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}} \text{WCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)))$ as follows:

$$\begin{aligned} &\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}, \mathbf{s}, \Lambda, \eta} \Lambda \\ \text{s.t. } &\Lambda \geq \max_{\mathbf{p} \in \mathcal{P}^{\text{Ellipsoidal}}} \eta + \frac{1}{1-\alpha} \mathbf{p}^T \mathbf{s}. \end{aligned} \quad (85)$$

Constraints (62)–(63).

Recall that the distribution probability $\mathbf{p}$ belongs to the ellipsoidal ambiguity set $\mathcal{P}^{\text{Ellipsoidal}}$ expressed in Eq. (36), the right-side hand of the first constraints in problem (85) can be expressed as

$$\max_{\mathbf{p} \in \mathcal{P}^{\text{Ellipsoidal}}} \eta + \frac{1}{1-\alpha} \mathbf{p}^T \mathbf{s} = \eta + \frac{1}{1-\alpha} (\mathbf{p}^0)^T \mathbf{s} + \frac{Y^*(\mathbf{s})}{1-\alpha}, \quad (86)$$

where $Y^*(\mathbf{s})$ denotes the optimal value of the following second-order cone programming.

$$\max_{\epsilon} \left\{ \mathbf{s}^T \mathbf{A}_1 \epsilon \mid \mathbf{e}^T \mathbf{A}_1 \epsilon = 0, -\mathbf{p}^0 - \mathbf{A}_1 \epsilon \leq 0, \|\epsilon\|_2 - 1 \leq 0 \right\}. \quad (87)$$

Similar to the methodology used to reformulate the expectation part, we have

$$Y^*(\mathbf{s}) = \min_{\hat{\boldsymbol{\vartheta}}, \hat{\theta}, \hat{\zeta}} \left\{ (\mathbf{p}^0)^T \hat{\boldsymbol{\vartheta}} + \hat{\zeta} \mid \begin{array}{l} \|\mathbf{A}_1^T \hat{\boldsymbol{\vartheta}} + \mathbf{A}_1^T \mathbf{s} - \hat{\theta} \mathbf{A}_1^T \mathbf{e}\|_2 \leq \hat{\zeta} \\ \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \mathbf{e} \eta \\ \hat{\boldsymbol{\vartheta}} \geq \mathbf{0}, \quad \hat{\zeta} \geq 0, \quad \mathbf{s} \geq 0. \end{array} \right\}. \quad (88)$$

Hence, we have the following minimisation problem.

$$\begin{aligned} &\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}, \mathbf{s}, \Lambda, \hat{\boldsymbol{\vartheta}}, \hat{\theta}, \hat{\zeta}, \hat{\eta}} \Lambda \\ \text{s.t. } &\Lambda \geq \eta + \frac{1}{1-\alpha} (\mathbf{p}^0)^T \mathbf{s} + \frac{1}{1-\alpha} [(\mathbf{p}^0)^T \hat{\boldsymbol{\vartheta}} + \hat{\zeta}], \\ &\|\mathbf{A}_1^T \hat{\boldsymbol{\vartheta}} + \mathbf{A}_1^T \mathbf{s} - \hat{\theta} \mathbf{A}_1^T \mathbf{e}\|_2 \leq \hat{\zeta}, \\ &\mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \mathbf{e} \eta, \\ &\hat{\boldsymbol{\vartheta}} \geq \mathbf{0}, \quad \hat{\zeta} \geq 0, \quad \mathbf{s} \geq 0. \end{aligned} \quad (89)$$

Following the logic of Proposition 1, we can prove the problem (85) is equivalent to problem (89).

Therefore, we can express the formula $\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \text{WMCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)))$ in the problem (32) as

$$\begin{aligned} & \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \text{WMCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))) \\ &= \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} (1 - \lambda) \sup_{\mathbf{p} \in \mathcal{P}^{\text{Ellipsoidal}}} \mathbb{E}(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))) + \lambda \text{WCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))) \\ &= \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \left\{ \min_{\theta, \theta, \zeta} (1 - \lambda) [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^T \mathbf{p}^0 + (\mathbf{p}^0)^T \boldsymbol{\theta} + \zeta] \mid \begin{array}{l} \|\mathbf{A}_1^T \boldsymbol{\theta} + \mathbf{A}_1^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \theta \mathbf{A}_1^T \mathbf{e}\|_2 \leq \zeta, \\ \boldsymbol{\theta} \geq \mathbf{0}, \quad \zeta \geq 0. \end{array} \right. \\ & \quad \left. + \lambda \min_{\hat{\theta}, \hat{\theta}, \hat{\zeta}, s, \eta} \left\{ \eta + \frac{1}{1-\alpha} (\mathbf{p}^0)^T \mathbf{s} + \frac{1}{1-\alpha} [(\mathbf{p}^0)^T \hat{\boldsymbol{\theta}} + \hat{\zeta}] \mid \begin{array}{l} \|\mathbf{A}_1^T \hat{\boldsymbol{\theta}} + \mathbf{A}_1^T \mathbf{s} - \hat{\theta} \mathbf{A}_1^T \mathbf{e}\|_2 \leq \hat{\zeta}, \\ \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \mathbf{e}\eta, \\ \hat{\boldsymbol{\theta}} \geq \mathbf{0}, \quad \hat{\zeta} \geq 0, \quad \mathbf{s} \geq \mathbf{0}. \end{array} \right\} \right\}. \end{aligned} \quad (90)$$

Finally, the proposed DRO model [DRP-DSRFLP] under the ellipsoidal ambiguity set $\mathcal{P}^{\text{Ellipsoidal}}$ can be equivalently reformulated as follows:

$$\begin{aligned} & \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi, \theta, \theta, \zeta, \hat{\theta}, \hat{\theta}, \hat{\zeta}, s, \eta} \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c^{\text{INV}} v_k + \sum_{l \in L} \sum_{d \in D} \sum_{k \in K} c_{kl}^{\text{MANA}} z_{idk} + \lambda \eta \\ & \quad + (1 - \lambda) [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^T \mathbf{p}^0 + (\mathbf{p}^0)^T \boldsymbol{\theta} + \zeta] + \frac{\lambda}{1-\alpha} [(\mathbf{p}^0)^T \mathbf{s} + (\mathbf{p}^0)^T \hat{\boldsymbol{\theta}} + \hat{\zeta}] \\ & \text{s.t. } \|\mathbf{A}_1^T \boldsymbol{\theta} + \mathbf{A}_1^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \theta \mathbf{A}_1^T \mathbf{e}\|_2 \leq \zeta, \\ & \quad \|\mathbf{A}_1^T \hat{\boldsymbol{\theta}} + \mathbf{A}_1^T \mathbf{s} - \hat{\theta} \mathbf{A}_1^T \mathbf{e}\|_2 \leq \hat{\zeta}, \\ & \quad \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \eta \mathbf{e}, \\ & \quad \boldsymbol{\theta} \geq \mathbf{0}, \quad \zeta \geq 0, \quad \hat{\boldsymbol{\theta}} \geq \mathbf{0}, \quad \hat{\zeta} \geq 0, \quad \mathbf{s} \geq \mathbf{0}, \\ & \quad \text{Constraints (2)–(13), Constraints (15)–(22)}. \end{aligned} \quad (91)$$

where $(\theta, \theta, \zeta, \hat{\theta}, \hat{\theta}, \hat{\zeta}) \in \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+^{|\mathcal{I}|} \times \mathbb{R} \times \mathbb{R}_+$ are Lagrange multipliers.

The proof of Theorem 2 is complete. $\square$

Appendix E. Proof of Theorem 3

Proof of Theorem 3. If the probability distribution of the discrete uncertain demand belongs to a polyhedral ambiguity set $\mathcal{P}^{\text{Polyhedral}}$ that is defined in Eq. (37), the expectation part of the model (34) can be reformulated as

$$\begin{aligned} & \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \max_{\mathbf{p} \in \mathcal{P}^{\text{Polyhedral}}} Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^T \mathbf{p} \\ &= \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \left\{ Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^T \mathbf{p}^0 + \max_{\epsilon} [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^T \mathbf{A}_2 \epsilon \mid \mathbf{e}^T \mathbf{A}_2 \epsilon = 0, \mathbf{p}^0 + \mathbf{A}_2 \epsilon \geq 0, \|\epsilon\|_1 \leq 1] \right\}. \end{aligned} \quad (92)$$

The Lagrange function of the inner maximisation problem can be expressed as follows:

$$\mathcal{L}(\boldsymbol{\pi}, \kappa, \varphi; \epsilon) = -Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^T \mathbf{A}_2 \epsilon + \boldsymbol{\pi}^T (-\mathbf{p}^0 - \mathbf{e}^T \mathbf{A}_2) \epsilon + \kappa (\mathbf{e}^T \mathbf{A}_2 \epsilon) + \varphi (\|\epsilon\|_1 - 1), \quad (93)$$

where $(\boldsymbol{\pi}, \kappa, \varphi) \in \mathbb{R}^{|\mathcal{I}|} \times \mathbb{R} \times \mathbb{R}$ are the Lagrange multipliers.

Then the Lagrange dual function can be formulated as

$$\begin{aligned} \mathcal{D}(\boldsymbol{\pi}, \kappa, \varphi; \epsilon) &= \min_{\epsilon} \mathcal{L}(\boldsymbol{\pi}, \kappa, \varphi; \epsilon) \\ &= -(\mathbf{p}^0)^T \boldsymbol{\pi} - \varphi - \max_{\epsilon} \{ [\mathbf{A}_2^T \boldsymbol{\pi} + \mathbf{A}_2^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \kappa \mathbf{A}_2^T \mathbf{e}]^T \epsilon - \varphi \|\epsilon\|_1 \} \\ &= -(\mathbf{p}^0)^T \boldsymbol{\pi} - \varphi - \mathcal{R}^* [\mathbf{A}_2^T \boldsymbol{\pi} + \mathbf{A}_2^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \kappa \mathbf{A}_2^T \mathbf{e}], \end{aligned} \quad (94)$$

where

$$\mathcal{R}^*(\cdot) = \begin{cases} 0 & \|\cdot\|_\infty \leq \varphi \\ \infty & \text{otherwise} \end{cases} \quad (95)$$

is the conjugate function of $\mathcal{R} = \varphi \|\epsilon\|_1$ (Boyd and Vandenberghe, 2004). For any $\varphi \geq 0$ and $\boldsymbol{\pi} \geq \mathbf{0}$, the Lagrange dual function can yield lower bounds. Referring the logic of the proof of Theorem 2, we have the equivalent problem of the expectation part as follows:

$$\begin{aligned} & \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi} \max_{\mathbf{p} \in \mathcal{P}^{\text{Polyhedral}}} Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^T \mathbf{p} \\ &= \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Psi, \boldsymbol{\pi}, \kappa, \varphi} \left\{ Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^T \mathbf{p}^0 + (\mathbf{p}^0)^T \boldsymbol{\pi} + \varphi \mid \begin{array}{l} \|\mathbf{A}_2^T \boldsymbol{\pi} + \mathbf{A}_2^T Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \kappa \mathbf{A}_2^T \mathbf{e}\|_\infty \leq \varphi \\ \boldsymbol{\pi} \geq \mathbf{0}, \quad \varphi \geq 0. \end{array} \right\}. \end{aligned} \quad (96)$$

Similar to the logic of Theorem 1, we can derive the equivalent problem of the $\min_{x,y,z,v,\Psi} \text{WCVaR}_\alpha(Q(x,y,z,v,\Theta(\xi)))$ as follows:

$$\begin{aligned} \min_{x,y,z,v,\Psi,s,\Lambda,\eta} \quad & \Lambda \\ \text{s.t.} \quad & \Lambda \geq \max_{\mathbf{p} \in \mathcal{P}^{\text{Polyhedral}}} \eta + \frac{1}{1-\alpha} \mathbf{p}^\top \mathbf{s}, \end{aligned} \quad (97)$$

Constraints (62)–(63).

Based on the logic of the proof of Theorem 2, we have the following equivalent minimisation problem.

$$\begin{aligned} \min_{x,y,z,v,\Psi,s,\Lambda,\hat{\pi},\hat{\varphi},\hat{\kappa},\eta} \quad & \Lambda \\ \text{s.t.} \quad & \Lambda \geq \eta + \frac{1}{1-\alpha} (\mathbf{p}^0)^\top \mathbf{s} + \frac{1}{1-\alpha} [(\mathbf{p}^0)^\top \hat{\pi} + \hat{\varphi}], \\ & \|\mathbf{A}_2^\top \hat{\pi} + \mathbf{A}_2^\top \mathbf{s} - \hat{\kappa} \mathbf{A}_2^\top \mathbf{e}\|_\infty \leq \hat{\varphi}, \\ & \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \mathbf{e}\eta, \\ & \hat{\pi} \geq \mathbf{0}, \quad \hat{\varphi} \geq 0, \quad \mathbf{s} \geq \mathbf{0}. \end{aligned} \quad (98)$$

Following the logic of Proposition 1, we can prove the problem (98) is equivalent to problem (97).

Therefore, we can express the formula $\min_{x,y,z,v,\Psi} \text{WMCVaR}_\alpha(Q(x,y,z,v,\Theta(\xi)))$ in the problem (32) as

$$\begin{aligned} & \min_{x,y,z,v,\Psi} \text{WMCVaR}_\alpha(Q(x,y,z,v,\Theta(\xi))) \\ &= \min_{x,y,z,v,\Psi} (1-\lambda) \sup_{\mathbf{p} \in \mathcal{P}^{\text{Polyhedral}}} \mathbb{E}(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))) + \lambda \text{WCVaR}_\alpha(Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))) \\ &= \min_{x,y,z,v,\Psi} \left\{ \min_{\pi,\kappa,\varphi} (1-\lambda) [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \pi + \varphi] \left| \begin{array}{l} \|\mathbf{A}_2^\top \pi + \mathbf{A}_2^\top Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \kappa \mathbf{A}_2^\top \mathbf{e}\|_\infty \\ \leq \varphi, \quad \pi \geq \mathbf{0}, \quad \varphi \geq 0. \end{array} \right. \right. \\ & \quad \left. \left. + \lambda \min_{\hat{\pi},\hat{\kappa},\hat{\varphi},s,\eta} \left\{ \eta + \frac{1}{1-\alpha} (\mathbf{p}^0)^\top \mathbf{s} + \frac{1}{1-\alpha} [(\mathbf{p}^0)^\top \hat{\pi} + \hat{\varphi}] \left| \begin{array}{l} \|\mathbf{A}_2^\top \hat{\pi} + \mathbf{A}_2^\top \mathbf{s} - \hat{\kappa} \mathbf{A}_2^\top \mathbf{e}\|_\infty \leq \hat{\varphi}, \\ \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \mathbf{e}\eta, \\ \hat{\pi} \geq \mathbf{0}, \quad \hat{\varphi} \geq 0, \quad \mathbf{s} \geq \mathbf{0}. \end{array} \right. \right\} \right\}. \end{aligned} \quad (99)$$

Finally, the proposed DRO model [DRP-DSRFLP] under the polyhedral ambiguity set $\mathcal{P}^{\text{Polyhedral}}$ can be equivalently reformulated as follows:

$$\begin{aligned} \min_{x,y,z,v,\Psi,\pi,\kappa,\varphi,\hat{\pi},\hat{\kappa},\hat{\varphi},s,\eta} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} c_k^{\text{INV}} v_k + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\ & + (1-\lambda) [Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \pi + \varphi] + \frac{\lambda}{1-\alpha} [(\mathbf{p}^0)^\top \mathbf{s} + (\mathbf{p}^0)^\top \hat{\pi} + \hat{\varphi}] \\ \text{s.t.} \quad & \|\mathbf{A}_2^\top \pi + \mathbf{A}_2^\top Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \kappa \mathbf{A}_2^\top \mathbf{e}\|_\infty \leq \varphi, \\ & \|\mathbf{A}_2^\top \hat{\pi} + \mathbf{A}_2^\top \mathbf{s} - \hat{\kappa} \mathbf{A}_2^\top \mathbf{e}\|_\infty \leq \hat{\varphi}, \\ & \mathbf{s} \geq Q(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \eta \mathbf{e}, \\ & \pi \geq \mathbf{0}, \quad \varphi \geq 0, \quad \hat{\pi} \geq \mathbf{0}, \quad \hat{\varphi} \geq 0, \quad \mathbf{s} \geq \mathbf{0}, \\ & \text{Constraints (2)–(13), Constraints (15)–(22)}. \end{aligned} \quad (100)$$

where $(\pi, \kappa, \varphi, \hat{\pi}, \hat{\kappa}, \hat{\varphi}) \in \mathbb{R}_+^{|\mathcal{E}|} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+^{|\mathcal{E}|} \times \mathbb{R} \times \mathbb{R}_+$ are Lagrange multipliers.

The proof of Theorem 3 is complete. $\square$

Appendix F. DRO formulations of model extension considering multiple relief items

The DRO formulation under box ambiguity set is as follows:

$$\begin{aligned} \min_{x,y,z,v,\Psi,\tau,\varsigma,\varrho,\hat{\tau},\hat{\varsigma},\hat{\varrho},s,\eta} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} \sum_{b \in B} c_b^{\text{INV}} v_{kb} + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\ & + (1-\lambda) [U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi))^\top \mathbf{p}^0 + \boldsymbol{\Omega}^\top \varsigma + \boldsymbol{\Omega}^\top \varrho] + \frac{\lambda}{1-\alpha} [(\mathbf{p}^0)^\top \mathbf{s} + \boldsymbol{\Omega}^\top \hat{\varsigma} + \boldsymbol{\Omega}^\top \hat{\varrho}] \\ \text{s.t.} \quad & \mathbf{e}\tau + \varsigma - \varrho = U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)), \\ & \mathbf{e}\hat{\tau} + \hat{\varsigma} - \hat{\varrho} = \mathbf{s}, \\ & \mathbf{s} \geq U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \Theta(\xi)) - \eta \mathbf{e}, \\ & \varsigma \geq \mathbf{0}, \varrho \geq \mathbf{0}, \hat{\varsigma} \geq \mathbf{0}, \hat{\varrho} \geq \mathbf{0}, \mathbf{s} \geq \mathbf{0}, \\ & \text{Constraints (2)–(5), (7)–(12), (43)–(44) and (46)–(53),} \end{aligned} \quad (101)$$

where $(\tau, \varsigma, \varrho, \hat{\tau}, \hat{\varsigma}, \hat{\varrho}) \in \mathbb{R} \times \mathbb{R}_+^{|\mathcal{E}|} \times \mathbb{R}_+^{|\mathcal{E}|} \times \mathbb{R} \times \mathbb{R}_+^{|\mathcal{E}|} \times \mathbb{R}_+^{|\mathcal{E}|}$ are dual variables.

The DRO formulation under ellipsoidal ambiguity set is as follows:

$$\begin{aligned}
\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}, \boldsymbol{\theta}, \boldsymbol{\zeta}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\zeta}}, \boldsymbol{s}, \boldsymbol{\eta}} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} \sum_{b \in B} c_b^{\text{INV}} v_{kb} + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\
& + (1 - \lambda) [U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \boldsymbol{\theta} + \boldsymbol{\zeta}] + \frac{\lambda}{1 - \alpha} [(\mathbf{p}^0)^\top \mathbf{s} + (\mathbf{p}^0)^\top \hat{\boldsymbol{\theta}} + \hat{\boldsymbol{\zeta}}] \\
\text{s.t.} \quad & \|\mathbf{A}_1^\top \boldsymbol{\theta} + \mathbf{A}_1^\top U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\theta} \mathbf{A}_1^\top \mathbf{e}\|_2 \leq \boldsymbol{\zeta}, \\
& \|\mathbf{A}_1^\top \hat{\boldsymbol{\theta}} + \mathbf{A}_1^\top \mathbf{s} - \hat{\boldsymbol{\theta}} \mathbf{A}_1^\top \mathbf{e}\|_2 \leq \hat{\boldsymbol{\zeta}}, \\
& \mathbf{s} \geq U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\eta} \mathbf{e}, \\
& \boldsymbol{\theta} \geq \mathbf{0}, \quad \boldsymbol{\zeta} \geq \mathbf{0}, \quad \hat{\boldsymbol{\theta}} \geq \mathbf{0}, \quad \hat{\boldsymbol{\zeta}} \geq \mathbf{0}, \quad \mathbf{s} \geq \mathbf{0}, \\
& \text{Constraints (2)–(5), (7)–(12), (43)–(44) and (46)–(53),}
\end{aligned} \tag{102}$$

where $(\boldsymbol{\theta}, \boldsymbol{\zeta}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\zeta}}) \in \mathbb{R}_+^{|\Xi|} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+^{|\Xi|} \times \mathbb{R} \times \mathbb{R}_+$ are Lagrange multipliers.

The DRO formulation under polyhedral ambiguity set is as follows:

$$\begin{aligned}
\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}, \boldsymbol{\pi}, \boldsymbol{\kappa}, \boldsymbol{\varphi}, \hat{\boldsymbol{\pi}}, \hat{\boldsymbol{\kappa}}, \hat{\boldsymbol{\varphi}}, \boldsymbol{s}, \boldsymbol{\eta}} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} \sum_{b \in B} c_b^{\text{INV}} v_{kb} + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\
& + (1 - \lambda) [U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \boldsymbol{\pi} + \boldsymbol{\varphi}] + \frac{\lambda}{1 - \alpha} [(\mathbf{p}^0)^\top \mathbf{s} + (\mathbf{p}^0)^\top \hat{\boldsymbol{\pi}} + \hat{\boldsymbol{\varphi}}] \\
\text{s.t.} \quad & \|\mathbf{A}_2^\top \boldsymbol{\pi} + \mathbf{A}_2^\top U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\kappa} \mathbf{A}_2^\top \mathbf{e}\|_\infty \leq \boldsymbol{\varphi}, \\
& \|\mathbf{A}_2^\top \hat{\boldsymbol{\pi}} + \mathbf{A}_2^\top \mathbf{s} - \hat{\boldsymbol{\kappa}} \mathbf{A}_2^\top \mathbf{e}\|_\infty \leq \hat{\boldsymbol{\varphi}}, \\
& \mathbf{s} \geq U_1(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\eta} \mathbf{e}, \\
& \boldsymbol{\pi} \geq \mathbf{0}, \quad \boldsymbol{\varphi} \geq \mathbf{0}, \quad \hat{\boldsymbol{\pi}} \geq \mathbf{0}, \quad \hat{\boldsymbol{\varphi}} \geq \mathbf{0}, \quad \mathbf{s} \geq \mathbf{0}, \\
& \text{Constraints (2)–(5), (7)–(12), (43)–(44) and (46)–(53),}
\end{aligned} \tag{103}$$

where $(\boldsymbol{\pi}, \boldsymbol{\kappa}, \boldsymbol{\varphi}, \hat{\boldsymbol{\pi}}, \hat{\boldsymbol{\kappa}}, \hat{\boldsymbol{\varphi}}) \in \mathbb{R}_+^{|\Xi|} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+^{|\Xi|} \times \mathbb{R} \times \mathbb{R}_+$ are Lagrange multipliers.

Appendix G. DRO formulations of model extension considering equity constraint in priority setting

The DRO formulation under box ambiguity set is as follows:

$$\begin{aligned}
\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}, \boldsymbol{\tau}, \boldsymbol{\varsigma}, \boldsymbol{\rho}, \hat{\boldsymbol{\tau}}, \hat{\boldsymbol{\varsigma}}, \hat{\boldsymbol{\rho}}, \boldsymbol{s}, \boldsymbol{\eta}} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} \sum_{b \in B} c_b^{\text{INV}} v_{kb} + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\
& + (1 - \lambda) [U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))^\top \mathbf{p}^0 + \boldsymbol{\Omega}^\top \boldsymbol{\varsigma} + \boldsymbol{\Omega}^\top \boldsymbol{\rho}] + \frac{\lambda}{1 - \alpha} [(\mathbf{p}^0)^\top \mathbf{s} + \boldsymbol{\Omega}^\top \hat{\boldsymbol{\varsigma}} + \boldsymbol{\Omega}^\top \hat{\boldsymbol{\rho}}] \\
\text{s.t.} \quad & \boldsymbol{\epsilon} \boldsymbol{\tau} + \boldsymbol{\varsigma} - \boldsymbol{\rho} = U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)), \\
& \boldsymbol{\epsilon} \hat{\boldsymbol{\tau}} + \hat{\boldsymbol{\varsigma}} - \hat{\boldsymbol{\rho}} = \mathbf{s}, \\
& \mathbf{s} \geq U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\eta} \mathbf{e}, \\
& \boldsymbol{\varsigma} \geq \mathbf{0}, \boldsymbol{\rho} \geq \mathbf{0}, \hat{\boldsymbol{\varsigma}} \geq \mathbf{0}, \hat{\boldsymbol{\rho}} \geq \mathbf{0}, \mathbf{s} \geq \mathbf{0}, \\
& \text{Constraints (2)–(5), (7)–(12), (43)–(44), (46)–(53) and (56),}
\end{aligned} \tag{104}$$

where $(\boldsymbol{\tau}, \boldsymbol{\varsigma}, \boldsymbol{\rho}, \hat{\boldsymbol{\tau}}, \hat{\boldsymbol{\varsigma}}, \hat{\boldsymbol{\rho}}) \in \mathbb{R} \times \mathbb{R}_+^{|\Xi|} \times \mathbb{R}_+^{|\Xi|} \times \mathbb{R} \times \mathbb{R}_+^{|\Xi|} \times \mathbb{R}_+^{|\Xi|}$ are dual variables.

The DRO formulation under ellipsoidal ambiguity set is as follows:

$$\begin{aligned}
\min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Psi}, \boldsymbol{\theta}, \boldsymbol{\zeta}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\zeta}}, \boldsymbol{s}, \boldsymbol{\eta}} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} \sum_{b \in B} c_b^{\text{INV}} v_{kb} + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\
& + (1 - \lambda) [U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi))^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \boldsymbol{\theta} + \boldsymbol{\zeta}] + \frac{\lambda}{1 - \alpha} [(\mathbf{p}^0)^\top \mathbf{s} + (\mathbf{p}^0)^\top \hat{\boldsymbol{\theta}} + \hat{\boldsymbol{\zeta}}] \\
\text{s.t.} \quad & \|\mathbf{A}_1^\top \boldsymbol{\theta} + \mathbf{A}_1^\top U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\theta} \mathbf{A}_1^\top \mathbf{e}\|_2 \leq \boldsymbol{\zeta}, \\
& \|\mathbf{A}_1^\top \hat{\boldsymbol{\theta}} + \mathbf{A}_1^\top \mathbf{s} - \hat{\boldsymbol{\theta}} \mathbf{A}_1^\top \mathbf{e}\|_2 \leq \hat{\boldsymbol{\zeta}}, \\
& \mathbf{s} \geq U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\theta}(\xi)) - \boldsymbol{\eta} \mathbf{e}, \\
& \boldsymbol{\theta} \geq \mathbf{0}, \quad \boldsymbol{\zeta} \geq \mathbf{0}, \quad \hat{\boldsymbol{\theta}} \geq \mathbf{0}, \quad \hat{\boldsymbol{\zeta}} \geq \mathbf{0}, \quad \mathbf{s} \geq \mathbf{0}, \\
& \text{Constraints (2)–(5), (7)–(12), (43)–(44), (46)–(53) and (56),}
\end{aligned} \tag{105}$$

where $(\boldsymbol{\theta}, \boldsymbol{\zeta}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\zeta}}) \in \mathbb{R}_+^{|\Xi|} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+^{|\Xi|} \times \mathbb{R} \times \mathbb{R}_+$ are Lagrange multipliers.

The DRO formulation under polyhedral ambiguity set is as follows:

$$\begin{aligned}
 \min_{\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\pi}, \kappa, \varphi, \hat{\boldsymbol{\pi}}, \hat{\kappa}, \hat{\varphi}, \mathbf{s}, \eta} \quad & \sum_{k \in K} \sum_{l \in L} c_l^{\text{OPEN}} x_{kl} + \sum_{k \in K} \sum_{b \in B} c_b^{\text{INV}} v_{kb} + \sum_{i \in I} \sum_{d \in D} \sum_{k \in K} c_{ki}^{\text{MANA}} z_{idk} + \lambda \eta \\
 & + (1 - \lambda) [U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi))^T \mathbf{p}^0 + (\mathbf{p}^0)^T \boldsymbol{\pi} + \varphi] + \frac{\lambda}{1 - \alpha} [(\mathbf{p}^0)^T \mathbf{s} + (\mathbf{p}^0)^T \hat{\boldsymbol{\pi}} + \hat{\varphi}] \\
 \text{s.t.} \quad & \|\mathbf{A}_2^T \boldsymbol{\pi} + \mathbf{A}_2^T U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \kappa \mathbf{A}_2^T \mathbf{e}\|_{\infty} \leq \varphi, \\
 & \|\mathbf{A}_2^T \hat{\boldsymbol{\pi}} + \mathbf{A}_2^T \mathbf{s} - \hat{\kappa} \mathbf{A}_2^T \mathbf{e}\|_{\infty} \leq \hat{\varphi}, \\
 & \mathbf{s} \geq U_2(\mathbf{x}, \mathbf{y}, \mathbf{z}, \mathbf{v}, \boldsymbol{\Theta}(\xi)) - \eta \mathbf{e}, \\
 & \boldsymbol{\pi} \geq \mathbf{0}, \quad \varphi \geq 0, \quad \hat{\boldsymbol{\pi}} \geq \mathbf{0}, \quad \hat{\varphi} \geq 0, \quad \mathbf{s} \geq \mathbf{0}, \\
 & \text{Constraints (2)–(5), (7)–(12), (43)–(44), (46)–(53) and (56),}
 \end{aligned} \tag{106}$$

where $(\boldsymbol{\pi}, \kappa, \varphi, \hat{\boldsymbol{\pi}}, \hat{\kappa}, \hat{\varphi}) \in \mathbb{R}_+^{|\mathcal{E}|} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+^{|\mathcal{E}|} \times \mathbb{R} \times \mathbb{R}_+$ are Lagrange multipliers.

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